

DermoGPT: Open Weights and Open Data for Morphology-Grounded Dermatological Reasoning MLLMs

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Abstract

Multimodal Large Language Models (MLLMs) show promise for medical applications, yet progress in dermatology lags due to limited training data, narrow task coverage, and lack of clinically-grounded supervision that mirrors expert diagnostic workflows. We present a comprehensive framework to address these gaps. First, we introduce **DermoInstruct**, a large-scale morphology-anchored instruction corpus comprising 211,243 images and 772,675 trajectories across five task formats, capturing the complete diagnostic pipeline from morphological observation and clinical reasoning to final diagnosis. Second, we establish **DermoBench**, a rigorous benchmark evaluating 11 tasks across four clinical axes: *Morphology*, *Diagnosis*, *Reasoning*, and *Fairness*, including a challenging subset of 3,600 expert-verified open-ended instances and human performance baselines. Third, we develop **DermoGPT**, a dermatology reasoning MLLM trained via supervised fine-tuning followed by our Morphologically-Anchored Visual-Inference-Consistent (MAVIC) reinforcement learning objective, which enforces consistency between visual observations and diagnostic conclusions. At inference, we deploy Confidence-Consistency Test-time adaptation (CCT) for robust predictions. Experiments show DermoGPT significantly outperforms 16 representative baselines across all axes, achieving state-of-the-art performance while substantially narrowing the human-AI gap. *DermoInstruct*, *DermoBench* and *DermoGPT* will be made publicly available at <https://github.com/mendicant04/DermoGPT> upon acceptance.

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1 Introduction

Skin diseases impose a substantial global burden, yet specialist access remains limited (Hay et al., 2014). Dermatological diagnosis requires differentiating hundreds of fine-grained conditions across modalities via systematic clinical reasoning (Mogensen et al., 2008). While Multimodal Large Language Models (MLLMs) show promise (Comanici et al., 2025; Bai et al., 2025), existing medical MLLMs (Chen et al., 2024; Zhou et al., 2024a; Liu et al., 2025b) struggle with dermatology’s specialized requirements due to limited training data, narrow task scopes, and lack of **interpretable reasoning mechanisms** aligned with clinical practice.

As summarized in Table 1, current resources exhibit three systemic limitations hindering clinical viability. **First, insufficient scale and diversity:** Existing resources like DermaSynth (Yilmaz et al., 2025) and MM-Skin (Zeng et al., 2025) typically cover only 2–3 tasks with limited samples. This scarcity fails to capture the long-tail visual complexity of the hundreds of conditions, severely limiting generalization. **Second, limited task formulations:** Existing instruction data and benchmarks predominantly rely on close-ended Multiple-Choice Question Answering (MCQAs) (Yim et al., 2024), inadequate for evaluating open-ended generation and multi-step reasoning required in clinical consultations. **Third, ungrounded clinical reasoning:** Unlike end-to-end models (Yan et al., 2025c,b) that map pixels directly to labels, expert dermatologists adhere to a “morphology-first” paradigm, parsing lesion morphology attributes to construct reasoning chains before diagnosis (Mogensen et al., 2008; Errichetti and Stinco, 2016). Current datasets lack supervision for this *morphology* → *reasoning* → *diagnosis* trajectory, yielding ungrounded systems prone to hallucinations inconsistent with visual evidence.

Dataset / Benchmark	Type		Scale			Features			
	Bench.	Train	#Tasks	#Images	#VQA Pairs	Multi-modal	Morph. CoT	CoT	Fairness
SkinCon (Ren et al., 2024)	✗	✓	2	3,886	–	✗	✗	✗	✗
SkinCap (Zhou et al., 2024b)	✗	✓	1	4,000	–	✗	✗	✗	✗
SkinCaRe (Shen et al., 2024)	✗	✓	2	7,041	7,041	✗	✗	✓	✗
DermaSynth (Yilmaz et al., 2025)	✗	✓	2	45,205	92,020	✓	✗	✓	✗
MM-Skin (Zeng et al., 2025)	✗	✓	3	11,039	27,412	✓	✗	✗	✗
DermaVQA (Yim et al., 2024)	✓	✓	1	3,434	1,488	✓	✗	✗	✗
DermBench (Shen et al., 2025b)	✓	✗	1	4,000	4,500	✓	✗	✓	✗
DermoInstruct (Ours)	✗	✓	4	211,243	772,675	✓	✓	✓	✗
DermoBench (Ours)	✓	✗	11	12,371	33,999	✓	✓	✓	✓

Table 1: Comparison of instruction datasets and benchmarks for dermatology MLLMs. Our datasets significantly expand task diversity and introduce morphology-grounded chain-of-thought reasoning (**Morph. CoT**) and fairness evaluation, addressing key gaps in existing resources.

To address these gaps, we propose a holistic framework centered on morphology-grounded reasoning. We first introduce **DermoInstruct**, a large-scale **morphology-anchored instruction corpus** unifying 14 heterogeneous public datasets under a shared diagnostic ontology with 9 superclasses and 325 fine-grained subclasses. The dataset contains 211,243 images and 772,675 instruction trajectories spanning 5 task formats: *free-text morphological description*, *structured attribute generation*, *clinically grounded Chain-of-Thought reasoning*, *flat diagnosis*, and *multi-turn hierarchical diagnosis*. This structured diversity ensures the model learns the complete diagnostic trajectory from lesion observation to morphology extraction to diagnostic reasoning, rather than mere label prediction. We also establish **DermoBench**, a comprehensive evaluation suite with 11 tasks across 4 clinical axes: *Morphology*, *Diagnosis*, *Reasoning*, and *Fairness* (Figure 1 and Table 2). For rigorous evaluation, we constructed 3,600 open-ended instances from a 900-case core image set with line-by-line specialist revision to guarantee morphological fidelity and reasoning validity, providing “Gold Standard” ground truth. We also benchmarked expert dermatologist performance as a clinical ceiling, enabling precise quantification of the Human-AI gap.

Building on these resources, we develop **DermoGPT**, a dermatology-specialized MLLM initialized from Qwen3-VL-8B. The training proceeds through two phases. First, Supervised Fine-Tuning (SFT) on DermoInstruct establishes foundational diagnostic capabilities. Second, a novel Morphologically-Anchored Visual-Inference-Consistent (MAVIC) reward aligns the model with clinical reasoning trajectories. MAVIC utilizes Group Relative Policy Optimization (GRPO) (Shao et al., 2024) to penalize logical disconnects between generated visual morphology descriptions and diagnostic conclusions, enforcing

the “morphology-first” reasoning trajectory. At inference, a Confidence-Consistency Test-time adaptation (CCT) scheme aggregates predictions to improve generalization. DermoGPT significantly outperforms 16 baselines across all 11 tasks, particularly in morphology understanding and reasoning consistency, narrowing the Human-AI gap.

Our contributions are three-fold: (1) **DermoBench Benchmark**: The first unified suite evaluating the full clinical pipeline beyond MC-QAs for dermatology. Validated against an expert-verified core set and human baselines, it exposes systemic reliability gaps in current MLLMs. (2) **DermoInstruct Dataset**: The largest ontology-aware corpus unifying 14 sources into structured multi-task trajectories, providing the essential supervision for versatile, clinically-aligned reasoning. (3) **DermoGPT**: The first clinical-aligned reasoning MLLM in dermatology utilizing the MAVIC and CCT. This approach yields substantial improvements, significantly narrowing the human-AI gap in both diagnostic accuracy and reasoning.

2 Related Work

Dermatology MLLMs and Reasoning. The landscape of dermatology AI has evolved from closed-set classification (AISuwaidan, 2023; Yan et al., 2025c) to open-ended multimodal reasoning. Early works relied on discriminative model with limited label spaces (Yan et al., 2025a; Kawahara et al., 2018). Recently, specialized MLLMs such as SkinGPT-4 (Zhou et al., 2024a), SkinGPT-R1 (Shen et al., 2025a), and Skin-R1 (Liu et al., 2025b) have adapted general foundation models to dermatology via instruction tuning. While these models demonstrate improved dialogue capabilities, they typically treat diagnostic reasoning as a latent, black-box process. Unlike our **DermoGPT**, which enforces an explicit *Morphology* $\rightarrow$ *Reasoning* $\rightarrow$ *Diagnosis* workflow via concept bottleneck,

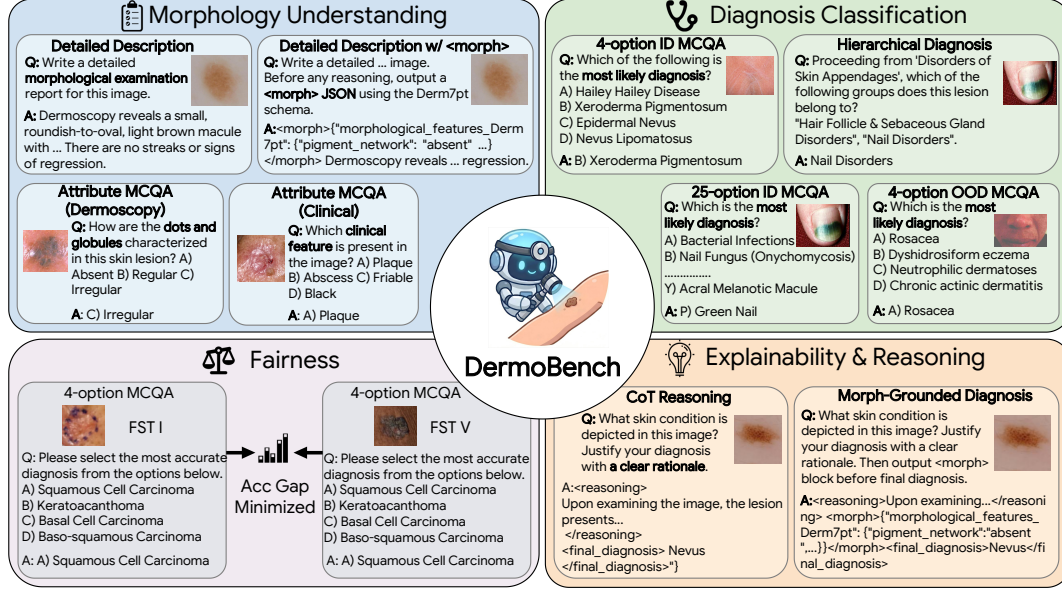


Figure 1: Overall architecture of DermoBench. DermoBench contains 11 subtasks spanning four axes: **Morphology** (Task 1.1 Detailed Description; Task 1.2 Morph-grounded Description; Task 1.3 Dermoscopic Attribute MCQA; Task 1.4 Clinical Attribute MCQA), **Diagnosis** (Task 2.1 4-option ID MCQA; Task 2.2 25-option ID MCQA; Task 2.3 hierarchical diagnosis; Task 2.4 4-option OOD MCQA), **Reasoning** (Task 3.1 CoT reasoning; Task 3.2 Morph-grounded Reasoning), and **Fairness** (Task 4). Note that the same set of images is used across all open-ended tasks (Tasks 1.1, 1.2, 3.1, and 3.2).

existing approaches lack fine-grained grounding, often leading to hallucinations where visual evidence contradicts diagnostic conclusions.

Dermatology Training Data and Benchmarks. The paradigm of dermatology AI has shifted from standard classification to large-scale vision-language alignment, exemplified by Derm1M (Yan et al., 2025a) and subsequent instruction-tuned MLLMs (Zhou et al., 2024a; Liu et al., 2025b). However, current approaches rely on small-scale instruction data with limited task diversity. Furthermore, evaluation remains underdeveloped—while DermBench (Shen et al., 2025b) assesses diagnostic narratives, it lacks rigorous workflow verification. To address these gaps, we introduce DermoInstruct, an expert-curated dataset with 772K morphology-grounded instruction pairs, and DermoBench, a multi-axis testbed that evaluates the full clinical workflow from morphology and diagnosis to OOD robustness and fairness.

3 DermoInstruct

To address the scarcity of clinically grounded training resources, we introduce DermoInstruct. Unlike prior works, this corpus is constructed to operationalize the “**morphology-first**” diagnostic workflow, providing high-quality supervision aligned with a unified ontology.

3.1 DermoInstruct Curation

The construction pipeline employs a four-step strategy to ensure both data scale and clinical rigor.

(1) Aggregation & Rigorous Cleaning: We aggregated 14 public datasets spanning clinical and dermoscopic modalities. To strictly prevent data leakage, we implemented a *patient-level split*. We further applied perceptual hashing (pHash, Hamming distance ≤ 2) to remove near-duplicate images, resulting in 211,243 distinct, high-quality images (see Appendix A for source details).

(2) Ontology Induction: Addressing the label fragmentation issue across heterogeneous sources, we employed GPT-5 to normalize 903 raw diagnostic strings into canonical clusters. These clusters were rigorously reviewed by two dermatologists to merge synonyms and resolve ambiguities, yielding a unified ontology of 9 superclasses and 325 fine-grained subclasses (Figure 2b; zoom in).

(3) Morphology-grounded Reasoning Synthesis: To transcend the limitations of naive CoT, we implemented a Clinically-Aligned Reasoning Synthesis pipeline that mirrors the expert diagnostic workflow: *Observation* $\rightarrow$ *Abstraction* $\rightarrow$ *Deduction*. We prompted Gemini-2.5-Flash (Comanici et al., 2025) via a strict dependency-aware protocol (detailed prompts could be found in Appendix C):

(i) **Morphological Inspection:** First, generate de-

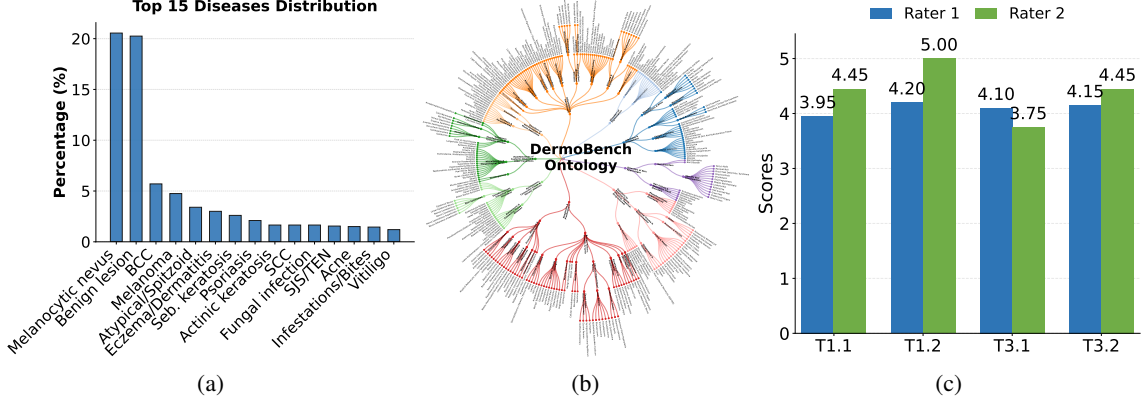


Figure 2: Overview of DermoBench. (a) Distribution of the top 15 diseases. (b) A unified ontology organizes 325 fine-grained diagnoses in DermoBench and DermoInstruct into 9 top-level super-classes. Zoom in for details. (c) Human ratings of LLM-as-a-Judge quality. 0 stands for “strongly disagree”, and 5 represents “strongly agree”

tailed descriptions of salient lesion structures (e.g., borders, symmetry) to simulate visual examination. (ii) **Schema-Based Anchoring**: Explicitly map these visual findings to standardized medical terminologies (seven-point checklist (Kawahara et al., 2018) for dermoscopy, general dermatology guidelines (Ren et al., 2024) for clinical images). This acts as a “concept bottleneck,” (Koh et al., 2020), anchoring pixel data to verifiable medical facts. (iii) **Evidence-Informed Diagnosis**: Finally, synthesize a reasoning chain that is rigorously conditioned on these extracted attributes. This enforces a reasoning trajectory where the model must justify the diagnosis via morphological evidences (e.g., “presence of atypical network implies higher risk of melanoma”), ensuring the reasoning is transparent, interpretable, and clinically coherent.

(4) **Diagnosis VQA Construction**: Complementing the open-ended reasoning, we leveraged the unified ontology to synthesize structured decision-making tasks that test the model’s diagnostic precision. For *Flat MCQAs*, we enforced clinical hardness by sampling distractors exclusively from sibling nodes or nearest neighbors (i.e., clinical mimics), demanding fine-grained discrimination beyond random guessing. For *Hierarchical Instructions*, we modeled diagnosis as a sequential root-to-leaf traversal with an adaptive correction mechanism: if the reasoning trajectory deviates, corrective prompts inject expert guidance to realign the diagnostic path, simulating the interactive pedagogy of medical training.

3.2 DermoInstruct Data Analysis

The final corpus comprises 211,243 multimodal images and 772,675 instructions (646k used for training after holding out DermoBench evaluation splits;

see Appendix A.2). As illustrated in Figure 2, the dataset features a realistic long-tail disease distribution (Fig. 2a) organized under our unified ontology of 9 superclasses and 325 subclasses (Fig. 2b). The instruction data across 4 major task dimensions spans 5 formats forming a complete diagnostic loop: (1) *Free-text morphological description*; (2) *Structured attribute generation* (for concept bottleneck training); (3) *Clinically grounded CoT reasoning*; (4) *Flat diagnosis*; and (5) *Multi-turn hierarchical diagnosis*. This structured diversity ensures the model learns to look, reason, and deduce, rather than just memorize labels.

4 DermoBench

4.1 Benchmark Construction

We construct DermoBench, a comprehensive evaluation suite comprising 33,999 VQA pairs spanning 11 subtasks across 4 dimensions: *Morphology*, *Diagnosis*, *Reasoning*, and *Fairness* (Table 2 and Appendix Figure 5). The benchmark consists of 3,600 open-ended instances from a 900-case core image set (enabling cross-task consistency evaluation across T1.1, T1.2, T3.1, T3.2) and 30,399 closed-ended MCQAs. Each open-ended sample underwent strict line-by-line dermatologist revision to serve as gold-standard references. Two reasoning tasks (T1.2, T3.2) require structured morphological evidence before diagnosis to prevent ungrounded predictions. Independent sanity checks by two dermatologists confirmed high annotation quality with mean scores of 3.88–4.60 in a 5-scale score across tasks (Appendix Figure 5b). The closed-ended component comprises 12,533 diagnoses, 654 fairness and 17,212 attribute-related MCQAs across 7 subtasks, including Out-of-Distribution tasks

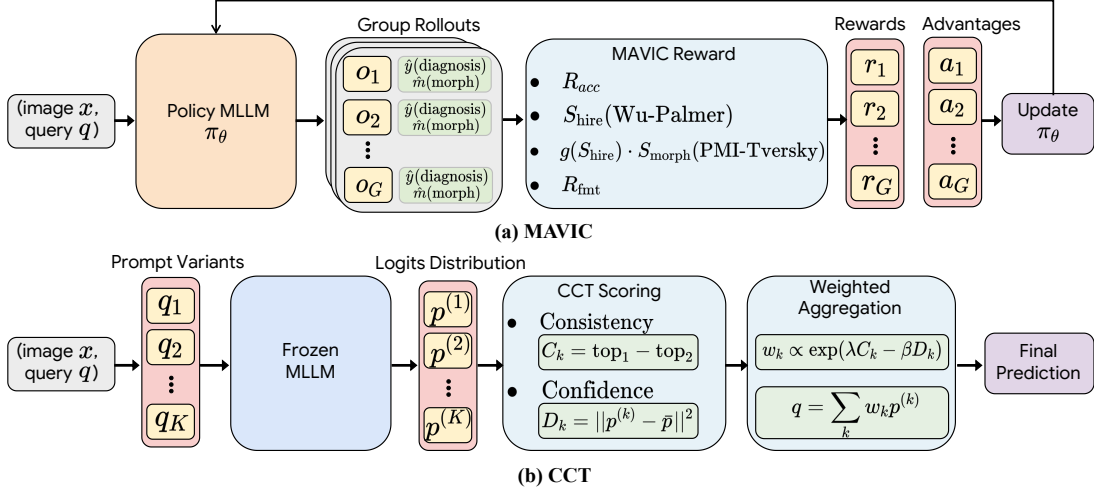


Figure 3: Method overview of MAVIC and CCT. (a) MAVIC integrates diagnosis accuracy, taxonomy-level similarity, gated morphology agreement, and format validity into a GRPO-style group reward to enforce morphology-first alignment. (b) CCT is a decoding-only test-time aggregation that reweights prompt-variant distributions by confidence and cross-variant consistency, requiring no parameter updates.

(T2.4). All images are isolated from training data to prevent leakage. Please refer to Appendix B for details about task definitions and data sources.

4.2 Evaluation Metrics

We adopt distinct metrics (Cai et al., 2025; Hao et al., 2025) tailored to the nature of each subtask. For closed-ended questions, we use standard accuracy. For open-ended tasks, we employ an LLM-as-a-Judge protocol (using Gemini-2.5-Pro), which compares model outputs against human-curated references to generate mean fidelity scores. Judge consistency was validated through model substitution experiments and human sanity checks. Crucially, to quantify the real-world utility gap, we invited board-certified dermatologists to complete all tasks. Their performance serves as the clinical ceiling, allowing us to precisely measure where MLLMs fall short compared to human experts. Complete LLM-as-a-judge protocol are in Appendix D.

5 DermoGPT

We aim to develop models that follow the dermatological reasoning chain *morphology* $\rightarrow$ *reasoning* $\rightarrow$ *diagnosis* with explicitly verifiable intermediate steps. We propose MAVIC (Morphologically-Anchored Visual-Inference-Consistent) reward, an end-to-end computable reward function requiring no external judge, achieving morphology-first alignment during RL training. We further introduce CCT (Confidence–Consistency Test-time adaptation), a plug-and-play decoding strategy that enhances OOD generalization without fine-tuning.

Hyperparameters and implementation details are documented in Appendix E.

5.1 MAVIC: Morphologically-Anchored Visual-Inference-Consistent Reward

We begin with multi-task supervised fine-tuning (SFT) on DermoInstruct using Qwen3-VL-8B-Instruct (Bai et al., 2025). We optimize cross-entropy loss for 1 epoch with LoRA (rank 64, $\alpha = 64$, dropout 0.05) while freezing the LLM and training the vision tower and projector, obtaining DermoGPT-SFT. To enable automatic verification of morphological evidence, we adopt a concept bottleneck framework (Koh et al., 2020) that compels the model to output structured morphological features following the “seven-point checklist” (Kawahara et al., 2018) and “general dermatology guideline” (Ren et al., 2024) schema. This structured output enables direct computation of morphology-level rewards without external judges—a key departure from prior approaches that rely on costly LLM-as-a-judge pipelines.

However, RL training for open-ended morphology descriptions faces a critical challenge: lack of directly verifiable reward signals. Diagnosis-only rewards are sparse and encourage shortcut learning that bypasses morphological evidence. To address this, we design MAVIC reward with the following components. Given an image and instruction, we sample G completions from the current policy following GRPO (Shao et al., 2024) and compute the following reward components for each rollout:

(1) R_{acc} : Standard 0-1 reward for tasks with unique

Axis	Task	Type	#Pairs
Morphology	T1.1 Detailed Description	Open-ended	900
	T1.2 Morph-grounded Description	Open-ended	900
	T1.3 Dermoscopic attribute MCQA	MCQA	5,530
	T1.4 Clinical attribute MCQA	MCQA	11,682
Diagnosis	T2.1 ID 4-way MCQA	MCQA	2,000
	T2.2 ID 25-way MCQA	MCQA	2,000
	T2.3 Hierarchical diagnosis	MCQA (multi-step)	2,000
	T2.4 OOD 4-way MCQA	MCQA	6,533
Reasoning	T3.1 CoT reasoning	Open-ended	900
	T3.2 Morph-grounded reasoning	Open-ended	900
Fairness	T4 Skin-type fairness MCQA	MCQA	654

Table 2: DermoBench tasks, sizes, and data sources.

ground-truth (e.g., MCQAs).

(2) $S_{\text{hier}} \in [0, 1]$: Hierarchical similarity over the diagnostic ontology using Wu-Palmer function (Wattiheluw and Sarno, 2018). This differentiates completely incorrect diagnoses from predictions correct at superclass level, mitigating sparse rewards while encouraging coarse-to-fine diagnostic alignment.

(3) $S_{\text{morph}} \in [0, 1]$: Morphology similarity computed via PMI-weighted Tversky matching on structured outputs (Derm7pt/SkinCon attributes).

(4) **Gating** $g(\cdot)$ and R_{fmt} : To prevent models from exploiting template-style morphology outputs when diagnoses diverge from ground truth, we progressively unlock morphology rewards only when diagnostic alignment is reasonable:

$$g(S_{\text{hier}}) = \sigma(k \cdot (S_{\text{hier}} - \mu)), \quad (1)$$

where μ is the median S_{hier} within each batch (adaptive difficulty threshold). R_{fmt} verifies JSON schema validity and critical tags to ensure auditable outputs. The total MAVIC reward is:

$$R = R_{\text{acc}} + \lambda_{\text{hier}} S_{\text{hier}} + \lambda_{\text{morph}} g(S_{\text{hier}}) S_{\text{morph}} + R_{\text{fmt}}, \quad (2)$$

with $\lambda_{\text{hier}} = \lambda_{\text{morph}} = 1$ by default. We optimize the standard GRPO objective using MAVIC rewards to obtain DermoGPT-RL. Complete implementation detail is in Appendix E.2.

5.2 Confidence–Consistency Test-time Adaptation

To further improve generalization under distribution shifts, we note that trivial deterministic decoding often yields unstable predictions on out-of-distribution (OOD) samples, yet full test-time fine-tuning is infeasible in clinical workflows. Thus, we propose CCT, a purely decoding-level strategy that enhances OOD robustness through weighted aggregation of multiple stochastic rollouts, without updating model parameters. The key insight is that reliable predictions should be both confident and

consistent across sampling variations, aligning with dermatological practice where diagnostic certainty requires stable evidence.

5.2.1 Confidence–Consistency Ensemble

At each decoding step t for input (x, query) , we sample K rollouts yielding token distributions $p_t^{(1)}, \dots, p_t^{(K)} \in \Delta^{V-1}$. For each rollout r , we compute:

Confidence C_r (**margin-based**): Let $p_{t,(1)}^{(r)}$ and $p_{t,(2)}^{(r)}$ denote the highest and second-highest probabilities in $p_t^{(r)}$. We define $C_r = p_{t,(1)}^{(r)} - p_{t,(2)}^{(r)} \in [0, 1]$. A larger margin indicates a more confident prediction. For discrete answer tasks, we compute this over option tokens; for free-form generation, over the full vocabulary.

Consistency D_r (**deviation from barycenter**): We compute the empirical barycenter $\bar{p}_t = \frac{1}{K} \sum_{j=1}^K p_t^{(j)}$ and set $D_r = \frac{1}{2} \|p_t^{(r)} - \bar{p}_t\|_2^2$. Rollouts that deviate significantly from $\bar{p}_t$ (large D_r) are downweighted exponentially.

We construct the aggregated distribution via weighted combination:

$$q_t = \sum_{r=1}^K w_r p_t^{(r)}, \quad w_r = \frac{\exp(\lambda C_r - \beta D_r)}{\sum_{j=1}^K \exp(\lambda C_j - \beta D_j)}, \quad (3)$$

where λ and β control the relative importance of confidence and consistency. The weighting exponentially suppresses outlier rollouts (high D_r) while favoring confident predictions (high C_r), ensuring predictions are both stable and confident, critical for clinical reliability. The next token is sampled from q_t , and this process repeats for each step. In practice, we set $K = 8$ and $\lambda = \beta = 1.0$.

5.2.2 Theoretical Guarantee

To formalize the robustness of this weighting scheme, we establish the following guarantee under distribution contamination; full proofs appear in Appendix G.

Theorem 1 (Robustness of CCT, informal). *Let $\{p_t^{(r)}\}_{r=1}^K$ be sampled from a mixture where fraction $(1 - \varepsilon)$ comes from a “good” component concentrated near p_t^* , and fraction ε comes from an arbitrary “bad” component ($\varepsilon < \frac{1}{2}$). Under bounded variance assumptions, there exist constants $\varepsilon_{\text{eff}}, C_U, \gamma_{\text{eff}} > 0$ such that:*

$$\|q_t - p_t^*\|_2 \leq \varepsilon_{\text{eff}} + C_U + \text{const} \cdot \exp(-\beta \gamma_{\text{eff}} + \lambda). \quad (4)$$

The bound shows that corrupted rollouts’ influence decays exponentially with β , keeping q_t

Model	Params	Task 1: Morphology					Task 2: Diagnosis								Task 3: Reasoning			Task 4	
		T1.1	T1.2	T1.3	T1.4	Avg.	In-Distribution (ID)				Out-of-Distribution (OOD)				T3.1	T3.2	Avg.	Fair.	
		(Desc)	(Struct)	(D7pt)	(SkinCon)	(T1)	4-clcs	25-clcs	Hier.	Avg.	DermlM	DDI	D7pt	SNU	Avg.	(CoT)	(M-CoT)	(T3)	(Score)
General Purpose MLLMs																			
GPT-4o-mini	/	34.55	51.80	41.19	61.09	47.16	59.50	34.75	65.90	53.38	52.12	58.54	56.48	59.17	56.57	42.83	51.65	47.24	94.06
Claude-Sonnet-4.5-Thinking	/	36.75	55.90	29.73	59.20	45.40	55.35	34.15	63.40	50.97	53.64	52.90	50.40	68.75	56.42	43.54	54.37	48.95	91.40
Gemini-2.5-Flash	/	40.08	53.48	39.28	66.59	49.86	72.60	47.20	70.31	63.37	66.33	59.15	53.96	65.42	61.21	48.92	58.49	53.70	79.89
GLM-4.5V	106B	36.85	42.75	45.50	52.03	44.28	63.65	28.85	52.39	48.30	45.51	48.17	43.08	57.08	48.46	44.19	53.28	48.73	93.59
Qwen2.5-VL-72B	72B	27.97	49.35	52.91	60.51	47.69	61.50	35.95	53.93	50.46	54.63	54.88	58.36	66.67	58.63	40.39	49.71	45.05	97.32
QVQ-72B-Preview	72B	22.38	41.02	49.77	59.20	43.09	64.65	47.30	57.25	56.40	60.53	53.66	56.92	62.92	58.51	51.56	54.14	52.85	86.26
Llama-3.2-90B	90B	28.20	44.43	35.84	49.19	39.41	47.85	51.65	51.20	50.23	44.76	49.09	37.14	49.58	45.14	44.61	56.14	50.38	91.31
Llama-3.2-11B	11B	12.33	38.48	39.13	29.93	29.97	29.25	16.50	35.98	27.58	25.50	21.80	26.90	42.92	29.28	36.16	38.29	37.22	53.85
Nemotron-Nano	12B	18.93	29.09	38.72	59.20	36.49	47.25	25.60	40.17	37.67	44.12	39.48	36.84	52.08	43.14	31.90	37.40	34.65	92.40
Qwen3-VL-32B	32B	50.30	57.43	46.15	60.67	53.64	64.25	38.05	64.08	55.46	48.13	57.93	63.11	69.58	59.69	55.04	53.85	54.45	81.78
Qwen3-VL-8B (Base)	8B	33.18	46.05	40.43	62.06	45.43	67.20	45.35	44.77	52.44	52.67	51.07	59.10	55.42	54.31	47.53	53.43	50.48	89.37
Medical/Dermatology Specialized																			
HuatuoGPT-Vis-7B	7B	18.15	34.50	33.82	38.15	31.15	51.60	26.05	46.10	41.25	31.40	36.13	41.64	47.92	39.27	39.41	43.98	41.69	76.80
LLaVA-Med-v1.5	7B	23.07	29.73	40.15	56.42	37.34	49.65	32.40	43.29	41.78	41.38	36.74	33.63	37.08	37.21	38.33	46.19	42.26	60.48
SkinVL-PubMM	7B	27.82	42.63	43.62	61.31	43.84	57.15	38.75	52.19	49.36	51.12	48.93	58.95	54.58	53.40	42.92	54.62	48.77	83.04
Lingshu-32B	32B	14.94	44.85	43.47	52.39	38.91	53.45	38.40	49.11	46.99	30.29	34.91	32.24	45.83	35.82	44.41	49.55	46.98	75.44
Lingshu-7B	7B	16.44	40.74	43.92	46.08	36.80	49.55	31.90	43.43	41.64	25.95	32.16	33.88	40.00	33.00	47.16	49.30	48.23	61.58
DermoGPT-SFT	8B	41.74	49.11	53.69	75.56	55.02	89.55	64.30	77.91	77.25	68.91	62.80	65.88	59.17	64.19	62.57	63.34	62.95	91.12
DermoGPT-SFT + CCT	8B	43.49	50.96	54.10	75.92	56.12	89.75	64.45	78.06	77.42	70.65	64.33	65.58	61.25	65.45	63.73	65.31	64.52	92.41
DermoGPT-RL	8B	43.93	59.29	56.53	76.67	59.10	90.30	64.60	79.12	78.01	69.68	62.80	68.59	60.00	65.27	66.04	65.48	65.76	93.49
DermoGPT-RL + CCT	8B	44.76	60.33	56.94	77.22	59.81	89.60	65.40	79.12	78.04	71.56	62.96	70.13	61.25	66.48	67.74	66.64	67.19	93.88
Human Performance	-	73.36	79.27	83.00	92.00	81.90	85.00	77.00	87.54	83.18	94.00	86.00	89.00	93.00	90.50	82.15	78.41	80.28	94.00

Table 3: **Main Results on DermoBench.** We evaluate models across four dimensions, and report each model’s parameter count when publicly available (**Params**; “/” denotes unknown). Blue columns indicate open-ended generation tasks (description and structured output), while orange columns indicate close-ended classification/scoring tasks. White columns represent aggregate metrics. **CCT** denotes our confidence-consistency test-time adaptation module. **Bold** indicates the best result in each column.

near p_t^* when β is sufficiently large relative to λ . This theoretical guarantee explains why CCT remains robust even when a substantial fraction (up to $\varepsilon < 50\%$) of rollouts are corrupted by distribution shifts—the aggregation automatically suppresses outliers without requiring knowledge of the corruption distribution.

6 Experiments

Performance on Closed-Ended Tasks. We evaluate model accuracy across Dermoscopic/Clinical Attribute Recognition (T1.3–1.4), Diagnosis (including In-Distribution 4-cl/25-cl/Hierarchical MCQA and OOD MCQA; T2), and Fairness (T4). Results demonstrate that our DermoGPT-SFT baseline alone establishes a new state-of-the-art, validating the high quality of our instruction data. On In-Distribution (ID) diagnosis (T2 Avg), SFT achieves 77.25%, surpassing its base model (Qwen3-VL-8B; 52.44%) and the strongest commercial baseline Gemini-2.5-Flash (63.37%) by substantial margins; notably, it excels in Hierarchical Diagnosis (77.91% vs. Gemini 70.31%) and Clinical Attribute Recognition (T1.4: 75.56% vs. Gemini 66.59%). Building on this foundation, our subsequent modules steadily improve robustness: the RL stage enhances OOD performance from 64.19% (SFT) to 65.27%, and the CCT module further elevates it to 66.48% by mitigating domain shifts. Consequently, our final DermoGPT-RL+CCT establishes a comprehensive new state-of-the-art, significantly

outperforming Gemini-2.5-Flash across all axes: it improves ID and OOD diagnostic accuracy by +14.67% and +5.27%, respectively; crucially, it simultaneously achieves an exceptional Fairness score of 93.88 (Task 4), surpassing Gemini (79.89) by nearly 14%, effectively minimizing diagnostic disparities across diverse skin tones.

Open-Ended Morphology & Reasoning. In Open-Ended Morphology and Reasoning tasks (T1.1, T1.2, T3), DermoGPT-RL+CCT demonstrates superior generation quality over both general-purpose MLLMs and existing medical-specialized models. Notably, previous medical specialized models (e.g., HuatuoGPT-Vis-7B) score only 41.69% on the Reasoning axis, lower than most general MLLMs, suggesting that naive fine-tuning without morphological constraints produces “black-box” classifiers rather than genuine reasoning agents. In contrast, our model scores 67.19% on the Reasoning axis on average (T3), outperforming Gemini-2.5-Flash (53.70%) by over 13.49%; this verifies that our Concept Bottleneck design effectively reduces hallucination by grounding reasoning in explicit morphological evidence. Human sanity checks confirmed high reliability of LLM-Judge scoring ($> 4.0/5.0$, Figure 2c). Despite these algorithmic advances, a significant Human-AI gap persists, particularly in Detailed Description (T1.1; 73.36 vs 44.76), highlighting that capturing fine-grained visual nuances remains a critical chal-

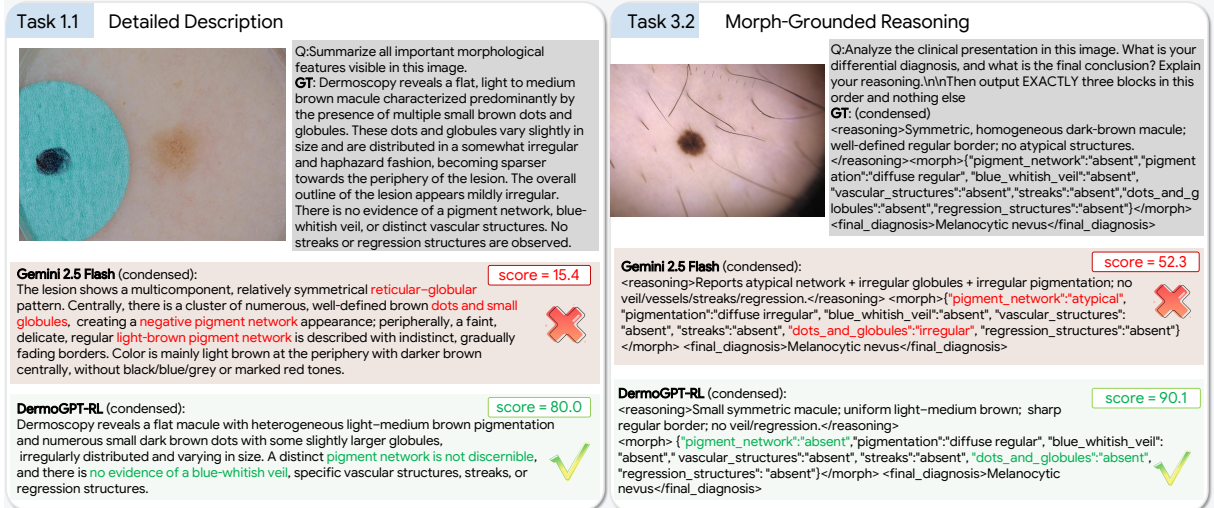


Figure 4: **Qualitative comparison on DermoBench.** Left: Task 1.1 (Detailed Description). Right: Task 3.2 (Morph-Grounded Reasoning with ultra-short structured outputs). Compared to Gemini-2.5 Flash, DerMoGPT-RL better matches the reference morphology and achieves higher scores.

Setting	T1.1	T1.2	T3.1	T3.2
SFT only	41.74	49.11	62.57	63.34
GRPO (acc+fmt)	35.13	41.20	61.34	59.88
w/o S_{morph}	39.65	48.09	65.40	65.27
w/o S_{hier}	42.59	50.11	63.96	65.02
w/o gate ($g=1$)	43.26	56.03	66.71	63.89
PMI→uniform	42.56	56.98	57.32	56.64
Full MAVIC	43.93	59.29	66.04	65.48

Table 4: MAVIC ablations under GRPO setup ($K=8$). Higher is better for all metrics.

Method	ID MCQA	OOD MCQA	Hier.	Fair.
Single ($K=1$)	77.80	65.27	79.63	93.49
Vote ($K=4$)	78.10	65.83	79.15	93.50
MeanProb ($K=4$)	77.95	65.69	79.51	93.32
ConfOnly ($K=4$)	78.40	66.47	79.49	93.09
ConsOnly ($K=4$)	78.35	66.59	79.82	93.58
CC (Ours, $K=4$)	78.80	66.27	80.31	93.76

Table 5: Ablation of confidence-consistency components on 900-case core set. Higher is better for all metrics.

lence.

Ablation Study. We further dissect component contributions on the core set and OOD benchmarks. Please refer to Appendix F for more results.

(1) MAVIC Reward Analysis. We first investigate the necessity of morphology-guided rewards (Table 4). Naively applying standard RL with only accuracy and format rewards (GRPO(acc+fmt)) proves detrimental, degrading performance below SFT baseline across all reasoning tasks. This indicates that unconstrained RL encourages metric gaming rather than genuine clinical reasoning. Incorporating morphological similarity (S_{morph}) and hierarchical diagnosis rewards (S_{hier}) steadily im-

proves performance. Crucially, the full MAVIC setup with gating mechanism (g) achieves peak performance (65.48 on T3.2). Ablating the gate drops performance to 63.89, confirming that difficulty-aware gating prevents the model from bypassing morphological evidence to make uninformed diagnostic guesses.

(2) CCT Test-Time Adaptation Analysis. We evaluate test-time inference with K prompt variants and find that Confidence-Consistency (CC) aggregation consistently outperforms standard ensemble baselines (Majority Vote, MeanProb). As shown in Table 5, on Task 2.1, neither signal alone is sufficient: ConfOnly (78.40%) and ConsOnly (78.35%) both underperform CC (78.80%), indicating complementary robustness cues. We also observe test-time scaling: as K increases from 2 to 8, OOD performance rises from 65.82% to 66.48%, supporting TTS for improved reliability.

Qualitative Analysis. Fig. 4 validates DerMoGPT’s reasoning superiority over Gemini-2.5-Flash, which exhibits hallucinated morphology concepts (Task 1.1) and inconsistent reasoning between observations and diagnoses (Task 3.2). MAVIC-guided training enables DerMoGPT to maintain strict alignment, achieving significantly higher accuracy in feature description and diagnostic consistency.

7 Conclusion

We present a comprehensive framework for dermatology MLLMs grounded in morphology-first clinical reasoning. Our unified data-benchmark-model

suite—comprising DermoInstruct, DermoBench, and DermoGPT—enables systematic training and evaluation across diverse dermatological tasks, significantly advancing the state-of-the-art while narrowing the human–AI performance gap. This work establishes a foundation for developing clinically-viable dermatology AI systems that mirror expert diagnostic workflows.

Limitations

Despite substantial progress, several limitations warrant discussion. First, while DermoGPT significantly narrows the human–AI gap, performance disparities persist across all tasks, highlighting the inherent difficulty of clinical-grade diagnostic reasoning. Second, although our benchmark is comprehensive, it may not fully capture the complexity of real-world clinical scenarios, such as patient-level holistic analysis (Yan et al., 2025c) or cases requiring longitudinal patient histories. Third, despite integrating expert knowledge during data curation, the morphology-grounded reasoning chains remain susceptible to noise, particularly in ambiguous cases where visual features alone are insufficient for definitive diagnosis. Finally, computational constraints limited our exploration of larger model architectures and full parameter fine-tuning, both of which may further improve performance.

References

- Lulwah AlSuwaidan. 2023. Deep learning based classification of dermatological disorders. *Biomedical engineering and computational biology*, 14:11795972221138470.
- Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, and 1 others. 2025. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*.
- Zhenyang Cai, Jiaming Zhang, Junjie Zhao, Ziyi Zeng, Yanchao Li, Jingyi Liang, Junying Chen, Yunjin Yang, Jiajun You, Shuzhi Deng, and 1 others. 2025. Dentalgpt: Incentivizing multimodal complex reasoning in dentistry. *arXiv preprint arXiv:2512.11558*.
- Bill Cassidy, Connah Kendrick, Andrzej Brodzicki, Joanna Jaworek-Korjakowska, and Moi Hoon Yap. 2022. Analysis of the isic image datasets: Usage, benchmarks and recommendations. *Medical image analysis*, 75:102305.
- Junying Chen, Ruyi Ouyang, Anningzhe Gao, Shunian Chen, Guiming Hardy Chen, Xidong Wang, Ruifei Zhang, Zhenyang Cai, Ke Ji, Guangjun Yu, Xiang Wan, and Benyou Wang. 2024. [Huatuogpt-vision, towards injecting medical visual knowledge into multimodal llms at scale](#). *Preprint*, arXiv:2406.19280.
- Albert S Chiou, Jesutofunmi A Omiye, Haiwen Gui, Susan M Swetter, Justin M Ko, Brian Gastman, Joshua Arbesman, Zhuo Ran Cai, Olivier Gevaert, Christoph Sadée, and 1 others. 2025. Multimodal image dataset for ai-based skin cancer (midas) benchmarking. *NEJM AI*, 2(6):AIdbp2400732.
- Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, and 1 others. 2025. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*.
- Marc Combalia, Noel CF Codella, Veronica Rotemberg, Brian Helba, Veronica Vilaplana, Ofer Reiter, Cristina Carrera, Alicia Barreiro, Allan C Halpern, Susana Puig, and 1 others. 2019. Bcn20000: Dermoscopic lesions in the wild. *arXiv preprint arXiv:1908.02288*.
- Roxana Daneshjou, Kailas Vodrahalli, Roberto A Novoa, Melissa Jenkins, Weixin Liang, Veronica Rotemberg, Justin Ko, Susan M Swetter, Elizabeth E Bailey, Olivier Gevaert, and 1 others. 2022. Disparities in dermatology ai performance on a diverse, curated clinical image set. *Science advances*, 8(31):eabq6147.
- DermNet. 2023. [Dermnet](#). Accessed: 2023.
- Enzo Errichetti and Giuseppe Stinco. 2016. Dermoscopy in general dermatology: a practical overview. *Dermatology and therapy*, 6(4):471–507.
- Matteo Farina, Gianni Franchi, Giovanni Iacca, Massimiliano Mancini, and Elisa Ricci. 2024. Frustratingly easy test-time adaptation of vision-language models. *Advances in Neural Information Processing Systems*, 37:129062–129093.
- Philippe Gottfrois, Fabian Gröger, Faly Herizo Andriambololoniaina, Ludovic Amruthalingam, Alvaro Gonzalez-Jimenez, Christophe Hsu, Agnes Kessy, Simone Lionetti, Daudi Mavura, Wingston Ng’ambi, and 1 others. 2024. Passion for dermatology: Bridging the diversity gap with pigmented skin images from sub-saharan africa. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 703–712. Springer.
- Matthew Groh, Caleb Harris, Luis Soenksen, Felix Lau, Rachel Han, Aerin Kim, Arash Koochek, and Omar Badri. 2021. Evaluating deep neural networks trained on clinical images in dermatology with the fitzpatrick 17k dataset. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1820–1828.
- Zhihui Guo, Xin Man, Hui Xu, and Jie Shao. 2025. Lisa: A layer-wise integration and suppression approach

- for hallucination mitigation in multimodal large language models. *arXiv preprint arXiv:2507.19110*.
- Jing Hao, Yuci Liang, Lizhuo Lin, Yuxuan Fan, Wenkai Zhou, Kaixin Guo, Zanting Ye, Yanpeng Sun, Xinyu Zhang, Yanqi Yang, and 1 others. 2025. Oralgpt-omni: A versatile dental multimodal large language model. *arXiv preprint arXiv:2511.22055*.
- Roderick J Hay, Nicole E Johns, Hywel C Williams, Ian W Bolliger, Robert P Dellavalle, David J Margolis, Robin Marks, Luigi Naldi, Martin A Weinstock, Sarah K Wulf, and 1 others. 2014. The global burden of skin disease in 2010: an analysis of the prevalence and impact of skin conditions. *Journal of investigative dermatology*, 134(6):1527–1534.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, and 1 others. 2022. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3.
- Jeremy Kawahara, Sara Daneshvar, Giuseppe Argenziano, and Ghassan Hamarneh. 2018. Seven-point checklist and skin lesion classification using multi-task multimodal neural nets. *IEEE journal of biomedical and health informatics*, 23(2):538–546.
- Khushbu and Sharun Akter. 2024. [Skin disease classification dataset](#).
- Newton M Kinyanjui, Timothy Odonga, Celia Cintas, Noel CF Codella, Rameswar Panda, Prasanna Sattigeri, and Kush R Varshney. 2020. Fairness of classifiers across skin tones in dermatology. In *International conference on medical image computing and computer-assisted intervention*, pages 320–329. Springer.
- Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. 2020. Concept bottleneck models. In *International conference on machine learning*, pages 5338–5348. PMLR.
- Pravin R Kshirsagar, Hariprasath Manoharan, S Shitharth, Abdulrhman M Alshareef, Nabeel Albishry, and Praveen Kumar Balachandran. 2022. Deep learning approaches for prognosis of automated skin disease. *Life*, 12(3):426.
- Zihan Li, Diping Song, Zefeng Yang, Deming Wang, Fei Li, Xiulan Zhang, Paul E Kinahan, and Yu Qiao. 2025. Visionunite: A vision-language foundation model for ophthalmology enhanced with clinical knowledge. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- Bo Liu, Ke Zou, Li-Ming Zhan, Zexin Lu, Xiaoyu Dong, Yidi Chen, Chengqiang Xie, Jiannong Cao, Xiao-Ming Wu, and Huazhu Fu. 2025a. Gemex: A large-scale, groundable, and explainable medical vqa benchmark for chest x-ray diagnosis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 21310–21320.
- Shengyuan Liu, Boyun Zheng, Wenting Chen, Zhihao Peng, Zhenfei Yin, Jing Shao, Jiancong Hu, and Yixuan Yuan. 2023. Endobench: A comprehensive evaluation of multi-modal large language models for endoscopy analysis. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Zehao Liu, Wejieying Ren, Jipeng Zhang, Tianxiang Zhao, Jingxi Zhu, Xiaoting Li, and Vasant G Honavar. 2025b. Skin-r1: Toward trustworthy clinical reasoning for dermatological diagnosis. *arXiv preprint arXiv:2511.14900*.
- Mette Mogensen, Hanan A Morsy, Lars Thrane, and Gregor BE Jemec. 2008. Morphology and epidermal thickness of normal skin imaged by optical coherence tomography. *Dermatology*, 217(1):14–20.
- Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori B Hashimoto. 2025. s1: Simple test-time scaling. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 20286–20332.
- Stephanie S Noronha, Mayuri A Mehta, Dweepna Garg, Ketan Kotecha, and Ajith Abraham. 2023. Deep learning-based dermatological condition detection: A systematic review with recent methods, datasets, challenges, and future directions. *IEEE Access*, 11:140348–140381.
- Andre GC Pacheco, Gustavo R Lima, Amanda S Salomao, Breno Krohling, Igor P Biral, Gabriel G De Angelo, Fábio CR Alves Jr, José GM Esgario, Alana C Simora, Pedro BC Castro, and 1 others. 2020. Padufes-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones. *Data in brief*, 32:106221.
- Tschandl Philipp, Akay Nisa Bengü, Rosendahl Cliff, Rotemberg Veronica, Todorovska Verche, Weber Jochen, Wolber Anna Katharina, Müller Christoph, Kurtansky Nicholas, Halpern Allan, and 1 others. 2025. Milk10k: A hierarchical multimodal imaging learning toolkit for diagnosing pigmented and non-pigmented skin cancer and its simulators. *Journal of Investigative Dermatology*.
- Zhihang Ren, Yunqi Li, Xinyu Li, Xinrong Xie, Erik P Duhaime, Kathy Fang, Tapabrata Chakraborti, Yunhui Guo, Stella X Yu, and David Whitney. 2024. Skincon: Towards consensus for the uncertainty of skin cancer sub-typing through distribution regularized adaptive predictive sets (draps). In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 405–415. Springer.
- Jinghan Ru, Jun Tian, Chengwei Xiao, Jingjing Li, and Heng Tao Shen. 2023. Imbalanced open set domain adaptation via moving-threshold estimation and gradual alignment. *IEEE Transactions on Multimedia*, 26:2504–2514.

- Jinghan Ru, Yuxin Xie, Xianwei Zhuang, Yuguo Yin, Zhihui Guo, Zhiming Liu, Qianli Ren, and Yuexian Zou. 2025. Do we really have to filter out random noise in pre-training data for language models? *arXiv preprint arXiv:2502.06604*.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, and 1 others. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.
- Yuhao Shen, Jiahe Qian, Zhangtianyi Chen, Yuanhao He, and Juexiao Zhou. 2025a. Skingpt-r1: Adapter-only dual distillation for efficient dermatology reasoning. *arXiv preprint arXiv:2511.15242*.
- Yuhao Shen, Jiahe Qian, Shuping Zhang, Zhangtianyi Chen, Tao Lu, and Juexiao Zhou. 2025b. Towards trustworthy dermatology mllms: A benchmark and multimodal evaluator for diagnostic narratives. *arXiv preprint arXiv:2511.09195*.
- Yuhao Shen, Liyuan Sun, Yan Xu, Wenbin Liu, Shuping Zhang, Shawn Afvari, Zhongyi Han, Jiaoyan Song, Yongzhi Ji, Tao Lu, and 1 others. 2024. Skincare: A multimodal dermatology dataset annotated with medical caption and chain-of-thought reasoning. *arXiv e-prints*, pages arXiv–2405.
- Yuxuan Sun, Yunlong Zhang, Yixuan Si, Chenglu Zhu, Kai Zhang, Zhongyi Shui, Jingxiong Li, Xuan Gong, XINHENG LYU, Tao Lin, and Lin Yang. 2025. Pathgen-1.6m: 1.6 million pathology image-text pairs generation through multi-agent collaboration. In *The Thirteenth International Conference on Learning Representations*.
- Lexiang Tang, Xianwei Zhuang, Bang Yang, Zhiyuan Hu, Hongxiang Li, Lu Ma, Jinghan Ru, and Yuexian Zou. 2025. Not all tokens and heads are equally important: Dual-level attention intervention for hallucination mitigation. *arXiv preprint arXiv:2506.12609*.
- Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. 2018. The ham10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific data*, 5(1):1–9.
- Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. 2020. Tent: Fully test-time adaptation by entropy minimization. *arXiv preprint arXiv:2006.10726*.
- Juncheng Wang, Yilan Zhang, Fengying Xie, and Jie Liu. 2025a. Enhancing diagnosis of psoriasis and inflammatory skin diseases: A spatially aligned multimodal model integrating clinical and dermoscopic images. *Journal of Investigative Dermatology*.
- Zixin Wang, Yadan Luo, Liang Zheng, Zhuoxiao Chen, Sen Wang, and Zi Huang. 2025b. In search of lost online test-time adaptation: A survey. *International Journal of Computer Vision*, 133(3):1106–1139.
- Abbi Ward, Jimmy Li, Julie Wang, Sriram Lakshminarasimhan, Ashley Carrick, Bilson Campana, Jay Hartford, Pradeep K Sreenivasaiiah, Tiya Tiyasirisokchai, Sunny Virmani, and 1 others. 2024. Creating an empirical dermatology dataset through crowdsourcing with web search advertisements. *JAMA Network Open*, 7(11):e2446615–e2446615.
- Fadli Husein Wattiheluw and Riyanarto Sarno. 2018. Developing word sense disambiguation corpuses using word2vec and wu palmer for disambiguation. In *2018 International Seminar on Application for Technology of Information and Communication*, pages 244–248. IEEE.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, and 1 others. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837.
- Siyuan Yan, Ming Hu, Yiwen Jiang, Xieji Li, Hao Fei, Philipp Tschandl, Harald Kittler, and Zongyuan Ge. 2025a. Derm1m: A million-scale vision-language dataset aligned with clinical ontology knowledge for dermatology. *arXiv preprint arXiv:2503.14911*.
- Siyuan Yan, Xieji Li, Ming Hu, Yiwen Jiang, Zhen Yu, and Zongyuan Ge. 2025b. Make: Multi-aspect knowledge-enhanced vision-language pretraining for zero-shot dermatological assessment. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 369–379. Springer.
- Siyuan Yan, Zhen Yu, Clare Primiero, Cristina Vico-Alonso, Zhonghua Wang, Litao Yang, Philipp Tschandl, Ming Hu, Lie Ju, Gin Tan, and 1 others. 2025c. A multimodal vision foundation model for clinical dermatology. *Nature Medicine*, pages 1–12.
- Abdurrahim Yilmaz, Sirin Pekcan Yasar, Gulsum Gencoglan, and Burak Temelkuran. 2024. Derm12345: A large, multisource dermatoscopic skin lesion dataset with 40 subclasses. *Scientific Data*, 11(1):1302.
- Abdurrahim Yilmaz, Furkan Yuceyalcin, Ece Gokyayla, Donghee Choi, Ozan Erdem, Ali Anil Demircali, Rahmetullah Varol, Ufuk Gorkem Kirabali, Gulsum Gencoglan, Joram M Posma, and 1 others. 2025. Dermasynt: Rich synthetic image-text pairs using open access dermatology datasets. *arXiv preprint arXiv:2502.00196*.
- Wen-wai Yim, Yajuan Fu, Zhaoyi Sun, Asma Ben Abacha, Meliha Yetisgen, and Fei Xia. 2024. Dermavqa: A multilingual visual question answering dataset for dermatology. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 209–219. Springer.
- Yuguo Yin, Yuxin Xie, Wenyuan Yang, Dongchao Yang, Jinghan Ru, Xianwei Zhuang, Liming Liang, and Yuexian Zou. 2025. ATRI: Mitigating multilingual

audio text retrieval inconsistencies by reducing data distribution errors. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5491–5504, Vienna, Austria. Association for Computational Linguistics.

P Zaballo, LJ Del Pozo, G Argenziano, C Medina, F Lacarrubba, B Ferrer, JM Martin, A Llambrich, I Zalaudek, and J Bañuls. 2019. Dermoscopy of cutaneous smooth muscle neoplasms: a morphological study of 136 cases. *Journal of the European Academy of Dermatology and Venereology*, 33(4):693–699.

Wenqi Zeng, Yuqi Sun, Chenxi Ma, Weimin Tan, and Bo Yan. 2025. Mm-skin: Enhancing dermatology vision-language model with an image-text dataset derived from textbooks. In *Proceedings of the 33rd ACM International Conference on Multimedia*, pages 3769–3778.

Juexiao Zhou, Xiaonan He, Liyuan Sun, Jiannan Xu, Xiuying Chen, Yuetan Chu, Longxi Zhou, Xingyu Liao, Bin Zhang, Shawn Afvari, and 1 others. 2024a. Pre-trained multimodal large language model enhances dermatological diagnosis using skingpt-4. *Nature Communications*, 15(1):5649.

Juexiao Zhou, Liyuan Sun, Yan Xu, Wenbin Liu, Shawn Afvari, Zhongyi Han, Jiaoyan Song, Yongzhi Ji, Xiaonan He, and Xin Gao. 2024b. Skincap: A multimodal dermatology dataset annotated with rich medical captions. *arXiv preprint arXiv:2405.18004*.

Xianwei Zhuang, Yuxin Xie, Yufan Deng, Liming Liang, Jinghan Ru, Yuguo Yin, and Yuexian Zou. 2025. Vargpt: Unified understanding and generation in a visual autoregressive multimodal large language model. *arXiv preprint arXiv:2501.12327*.

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A Source Datasets and Extended Related Work

A.1 Source Dermatology Datasets

To construct DermoInstruct and DermoBench, we aggregate fourteen public or institutionally curated dermatology datasets covering clinical photographs, dermoscopic images, and smartphone or teledermatology photos from diverse healthcare systems: Daffodil (Khushbu and Akter, 2024), DermNet (DermNet, 2023), Fitzpatrick17k (Groh et al., 2021), ISIC Archive (Cassidy et al., 2022), MIDAS (Chiou et al., 2025), PAD-UFES-20 (Pacheco et al., 2020), PASSION (Gottfrois et al., 2024), PUMCH (Wang et al., 2025a), SCIN (Ward et al., 2024), SD-198 (Kinyanjui et al., 2020), BCN20000 (Combalia et al., 2019), HAM10000 (Tschandl et al., 2018), Derm12345 (Yilmaz et al., 2024), and MILK10k (Philipp et al., 2025). Note that images hosted on the ISIC platform that are not part of these named subsets are grouped into “ISIC Archive” collection. These datasets span pigmented and non-pigmented lesions, benign and malignant conditions, a wide range of anatomic sites and skin tones, and both controlled and real-world acquisition conditions. We briefly summarize their scope in Table 6; the main paper focuses on the unified ontology and task construction built on top of these sources.

Across these datasets, we harmonize heterogeneous diagnosis labels into a unified hierarchy of superclasses and subclasses, and map existing attribute schemas (e.g., dermoscopic structures, pigmentation patterns) into a common morphology ontology used consistently throughout DermoInstruct and DermoBench.

A.2 Leakage prevention and de-duplication.

We split data at the *patient* level (all images from the same `patient_id` are confined to a single split), allowing multiple cases per patient in the test set but ensuring no patient overlap with training. We exclude images from DDI, SCIN, PAD, SkinCon, and Derm7pt from training and reserve them for evaluation-only settings. Finally, we apply near-duplicate filtering with perceptual hashing (pHash; Hamming distance ≤ 2) to remove visually redundant images. In total, we retain 646,018 pairs for training after leakage controls and de-duplication.

A.3 Dermatology Benchmarks and Vision-Language Models

Traditional deep-learning systems for dermatology have focused on single-image diagnosis of a limited set of conditions, often trained and evaluated on individual datasets such as HAM10000 or ISIC, and commonly framed as closed-set classification tasks (Kshirsagar et al., 2022; AlSuwaidan, 2023; Noronha et al., 2023; Daneshjou et al., 2022). Recent work has begun to emphasize both fairness and robustness, highlighting disparities across skin tones and acquisition conditions and calling for more diverse benchmarks (Groh et al., 2021; Kinyanjui et al., 2020; Ward et al., 2024; Daneshjou et al., 2022).

In parallel, several multimodal and vision-language dermatology datasets and models have emerged. MAKE (Yan et al., 2025b) pre-trains a dermatology VLM with multi-aspect knowledge, and PanDerm (Yan et al., 2025c) proposes a dermatology vision foundation model trained on large-scale multimodal data. SkinGPT-4 (Zhou et al., 2024a), Skin-R1 (Liu et al., 2025b), and SkinGPT-R1 (Shen et al., 2025a) explore instruction-tuning and reasoning-style training for dermatology LLMs. DermBench (Shen et al., 2025b) and DermaVQA (Yim et al., 2024) provide evaluation datasets for diagnostic narratives and question answering, while SkinCap (Zhou et al., 2024b) and SkinCaRe (Shen et al., 2024) enrich image-text pairs with medical captions and chain-of-thought reasoning. More recently, Derm1M (Yan et al., 2025a) and DermaSynth (Yilmaz et al., 2025) scale dermatology vision-language data to the million-sample regime.

Beyond dermatology, there is a growing ecosystem of multimodal medical benchmarks and foundation models, such as GEMEX for chest X-ray VQA (Liu et al., 2025a), PathGen for pathology image-text pairs (Sun et al., 2025), EndoBench for endoscopy (Liu et al., 2023), and VisionUnite for ophthalmology (Li et al., 2025). Compared to these efforts, DermoBench is specifically designed to evaluate dermatology MLLMs along a morphology $\rightarrow$ reasoning $\rightarrow$ diagnosis axis, with fairness and robustness explicitly foregrounded.

A.4 Concept Bottleneck Models and Morphology-Grounded Reasoning

Concept bottleneck models (CBMs) explicitly insert an interpretable concept layer between raw

Dataset	Modality	Population / setting	Scale	Notes
Daffodil (Khushbu and Akter, 2024)	Dermoscopic	Bangladesh hospital	S	Biopsy-proven dermoscopy dataset.
DermNet (DermNet, 2023)	Clinical	Global web atlas	L	Expert-curated clinical photos.
Fitzpatrick17k (Groh et al., 2021)	Clinical	US outpatient clinics	M	Includes Fitzpatrick skin-type labels.
ISIC Archive (Casidy et al., 2022)	Dermoscopic	Multi-center dermoscopy	L	Standard benchmark for dermoscopic lesions.
MIDAS (Chiou et al., 2025)	Clinical & dermoscopic	Multi-institution NEJM AI dataset	M	Paired clinical/dermoscopy with biopsy labels.
PAD-UFES-20 (Pacheco et al., 2020)	Clinical	Brazilian teledermatology	S-M	Smartphone photos with rich metadata.
PASSION (Gottfrois et al., 2024)	Clinical	Sub-Saharan Africa	M	Smartphone images emphasizing pigmented skin.
PUMCH (Wang et al., 2025a)	Clinical	Chinese tertiary hospital	M	Broad inflammatory and neoplastic diseases.
SCIN (Ward et al., 2024)	Clinical	US crowdsourced users	M	Diverse smartphone photos with demographics.
SD-198 (Kinyanjui et al., 2020)	Clinical	China dermatology clinic	S-M	198-category long-tail dataset.
BCN20000 (Combalia et al., 2019)	Dermoscopic	Barcelona tertiary center	M-L	Large European dermoscopy cohort.
HAM10000 (Tschandl et al., 2018)	Dermoscopic	Austria & Australia	M	Classic dermoscopy benchmark.
Derm12345 (Yilmaz et al., 2024)	Dermoscopic	Turkish hospital	M	40-class dermoscopic dataset.
MILK10k (Philipp et al., 2025)	Clinical & dermoscopic	ISIC multimodal cohort	M	Paired clinical/dermoscopy with metadata.

Table 6: Summary of the fourteen source dermatology datasets used to construct DermoInstruct and DermoBench. “Scale” is qualitative (S: <5k images, M: 5k-20k, L: >20k).

features and task predictions: the model first predicts a vector of human-understandable concepts and then predicts the final label from those concepts (Koh et al., 2020). Such models allow users to inspect and intervene on the intermediate concept predictions, improving transparency and enabling richer human-model interaction. Subsequent work has studied robustness, intervention strategies, and automatic discovery of concepts, but the core idea remains to align model internals with domain-relevant abstractions.

Dermatology is naturally aligned with the CBM paradigm, because clinical practice is organized around lesion morphology. Dermatologists rely on structured morphology descriptors in both clinical and dermoscopic settings (Mogensen et al., 2008; Errichetti and Stinco, 2016; Zaballos et al., 2019), and recent datasets such as the SkinCon schema and the dermoscopic seven-point checklist

provide explicit morphology annotations for skin lesions (Ren et al., 2024; Kawahara et al., 2018). Our benchmark instantiates a *soft* concept bottleneck for dermatology: Task 1 evaluates morphology descriptions and attributes, Task 3 assesses chain-of-thought reasoning grounded in these concepts, and Task 2 measures whether diagnoses are consistent with both. Rather than inserting a fixed-dimensional concept layer into a single network, we expose morphology, reasoning, and diagnosis as separate but tightly coupled tasks, and exploit cross-task consistency as both a training signal (via DermoInstruct) and an evaluation criterion (via DermoBench).

A.5 Reinforcement Learning, GRPO, and Instruction-Tuned MLLMs

Reinforcement learning has become a central tool for enhancing the reasoning capabilities of large

language models beyond standard supervised fine-tuning. DeepSeekMath (Shao et al., 2024), for example, combines continued pre-training on math-heavy corpora with RL and introduces *Group Relative Policy Optimization* (GRPO), a variant of PPO that replaces a learned value-function critic with a group-based baseline over multiple sampled trajectories. GRPO-style objectives have quickly been adopted in reasoning-focused LLMs because they are sample-efficient, remove the need for a separate critic network, and work well with verifiable or heuristic reward signals.

Our MAVIC framework is inspired by this line of work but tailors the reward design to dermatology. Instead of rewarding only final correctness, we combine multiple terms capturing hierarchical diagnosis correctness, proximity in the ontology, morphology-grounded agreement with Task 1 outputs, and format constraints. This connects GRPO-style RL with clinical desiderata such as lesion understanding and cross-skin-type robustness, and is complementary to standard instruction-tuning with LoRA adaptation (Hu et al., 2022) and chain-of-thought prompting (Wei et al., 2022) used in general-purpose MLLMs such as Qwen3-VL and Gemini (Bai et al., 2025; Comanici et al., 2025; Tang et al., 2025) and in domain-specific models such as SkinGPT-4 and Skin-R1 (Zhou et al., 2024a; Liu et al., 2025b; Shen et al., 2025a; Zhuang et al., 2025; Guo et al., 2025).

A.6 Test-Time Adaptation and Test-Time Scaling

Test-time adaptation. Test-time adaptation (TTA) adapts a pre-trained model to unlabeled test data at deployment time, typically to mitigate covariate shifts without full re-training. Classical domain adaptation methods (Ru et al., 2023; Wang et al., 2020) update batch-normalization statistics or minimize prediction entropy, while more recent work explores online adaptation, pseudo-labeling, and robustness under dynamic streams (Wang et al., 2025b; Ru et al., 2025; Yin et al., 2025). For vision-language models, recent methods study both optimization based and optimization-free strategies. ZERO (Farina et al., 2024) shows that a surprisingly strong VLM TTA baseline can be obtained by aggressive test-time augmentation, temperature-0 prediction, and confidence-based marginalization, requiring only a single batched forward pass and no backpropagation. These results demonstrate that much of the benefit of

prompt-tuning style TTA can be captured by carefully designed test-time inference procedures.

Our CCT framework is complementary to these methods. Instead of updating model parameters, we adapt how the model is *queried* and how multiple stochastic predictions are aggregated: we sample multiple responses under morphology- and diagnosis-focused prompting, then aggregate them using confidence- and consistency-based weighting across tasks, images, and augmentations. This can be seen as a lightweight, domain-specific TTA scheme that relies on cross-task dermatology priors rather than parameter updates.

Test-time scaling. Test-time scaling (TTS) refers to improving model performance by allocating more compute at inference time without changing model parameters (Guo et al., 2025; Muennighoff et al., 2025). In the LLM literature, canonical examples include chain-of-thought prompting with self-consistency, where multiple reasoning paths are sampled and the majority answer is selected, and best-of- n sampling guided by task-specific scorers (Wei et al., 2022). Such techniques can substantially improve reasoning quality but incur linear cost in the number of samples.

Our CCT procedure can be interpreted as a specialized TTS scheme for dermatology MLLMs. By combining multi-sample decoding with confidence- and consistency-based aggregation across morphology, reasoning, and diagnosis tasks, CCT leverages the structure of DermoBench to stabilize predictions under distribution shifts (e.g., across devices or skin-tone groups) while keeping computation modest relative to naive best-of- n sampling.

B DermoBench Task Definitions and Data Sources

B.1 Task Overview and Sample Statistics

Table 2 summarizes all DermoBench subtasks, data sources, and sample sizes. The complete benchmark contains 33,999 VQA-style samples, distributed as follows: Task 1 has 19,012 samples; Task 2 has 12,533; Task 3 has 1,800; and Task 4 has 654.

B.2 Training Isolation and Leakage Control (Clean Separation)

To ensure credible evaluation results, DermoBench implements the following isolation strategies:

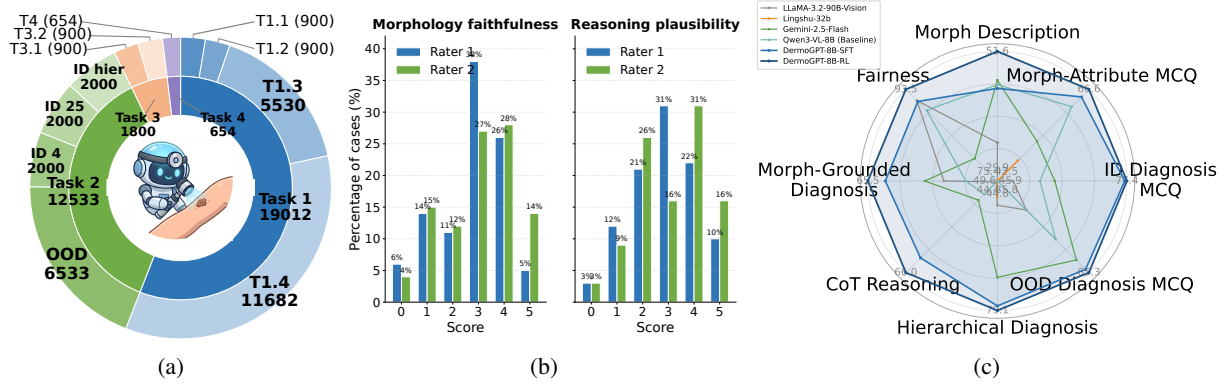


Figure 5: Benchmark statistics and key evaluation dimensions of DermoBench and DermoInstruct. (a) Task-wise and sub-task-wise distribution of VQA pairs. (b) Human ratings of synthesized morphological features and CoT of DermoInstruct. (c) Performance of representative MLLMs.

(1) Image-level Isolation. Unless explicitly stated, DermoBench images are sourced from datasets unused in DermoInstruct or from strictly held-out splits of the same source datasets. Critical morphology evaluation datasets such as Derm7pt and SkinCon are designated as evaluation-exclusive sources, with no images or labels utilized for training.

(2) Text-level Isolation. Reference texts for all open-ended tasks (T1.1/T1.2/T3.1/T3.2)—including morphological reports, attribute JSONs, reasoning chains, and diagnostic statements—are excluded from training corpora to prevent artificially inflated performance through answer memorization.

(3) Question/Template-level Isolation. Both multiple-choice and open-ended tasks employ minimal sets of semantically equivalent templates. We perform rigorous deduplication checks between training and evaluation template sets, and provide complete template inventories with cryptographic hashes for reproducibility upon release (see Appendix C).

B.3 Morphology Understanding (Task 1.x)

B.3.1 T1.1–T1.2: Open-Ended Morphology Evaluation on 900-Case Core Set

Input. A single clinical or dermoscopic image + instruction.

Output and Format Constraints.

- **T1.1 (Morph report):** Generate a structured morphological examination report covering key aspects including lesion type, color, border, surface/scales, and distribution.

- **T1.2 (Morph JSON + report):** In addition to the report, output a JSON object wrapped in `<morph>...</morph>` tags. Dermoscopic images follow Derm7pt checklist fields; clinical images follow SkinCon fields.

Gold Standard Construction (Core Process).

We first use a strong VLM to generate for each core set image: (i) morphological report, (ii) attribute JSON, and (iii) diagnostic reasoning with final diagnosis (for Task 3.x). Dermatologists then conduct line-by-line review and revision to ensure (a) textual descriptions align with visible evidence in images, (b) JSON field values conform to clinical terminology and definitions, and (c) consistency between descriptions and diagnoses. Detailed review guidelines, conflict resolution examples, and final consistency checks are provided in Appendix B.7.

B.3.2 T1.3: Dermoscopic Attribute MCQA

Data Source. The dermoscopic test split of Derm7pt is used for evaluation. Although Derm7pt provides training splits, we exclude all its images and labels from training.

Question Construction. Each question queries one attribute from the Derm7pt checklist (e.g., pigment network, streaks, etc.), with options corresponding to valid states for that attribute. Question templates and option generation rules are specified in Appendix C.

B.3.3 T1.4: SkinCon Attribute Multiple-Choice Questions (Clinical Attribute MCQA)

Data Source. SkinCon does not provide an official test split; we treat all its annotated samples as evaluation-only, generating MCQAs from its mor-

phological annotations. Question and option construction follow the same principles as above, with fields and value spaces determined by the SkinCon schema.

B.4 Diagnosis classification (Task 2.x)

B.4.1 In-distribution (ID) diagnosis (T2.1–T2.3)

Data sources and partitioning. The ID diagnosis evaluation set was constructed by extracting strictly held-out images from the same 14 source datasets as DermoInstruct (completely isolated from training instruction pairs, see Appendix B.2).

T2.1: 4-way MCQA (leaf-level). The correct option is a fine-grained leaf-node diagnosis; distractors are preferentially sampled from neighboring nodes/siblings under the same parent node in the unified ontology to enhance "clinical confusability."

T2.2: 25-way MCQA (coarse-grained triage). The 325 leaf-node diagnoses are collapsed into 25 coarse-grained categories with stronger clinical significance, creating a fixed option menu to simulate real-world triage scenarios.

T2.3: Hierarchical diagnosis. A single diagnosis is decomposed into sequential decisions along the ontology path (root→leaf). Each question corresponds to one step along the path, with both per-level accuracy and path-level metrics measured.

B.4.2 Out-of-distribution (OOD) diagnosis (T2.4)

Data sources. Evaluation partitions from multiple external dermoscopy/clinical datasets are used, including Derm1M educational split, Derm7pt, DDI, and SNU134.

Key setting: Non-aligned label spaces. Unlike ID tasks, OOD tasks construct MCQAs within each dataset’s **original label space**: We do not map ground-truth labels or options to a unified ontology. Consequently, models must simultaneously handle visual distribution shifts and label space mismatches, preventing inflated scores from "interpolating" on a unified taxonomy.

MCQA construction. For each sample, the original dataset label serves as the correct option; distractors are sampled from the same dataset’s label set (potentially weighted by class frequency or confusability).

B.5 Reasoning (Task 3.x)

B.5.1 T3.1: CoT reasoning

Data and objective. We use the same 900-case core set as in T1.1/T1.2. Models must output reasoning text enclosed in `<reasoning>...</reasoning>` tags, connecting visible evidence with candidate diagnoses, and provide the final diagnosis within `<final_diagnosis>` tags.

B.5.2 T3.2: Morph-grounded reasoning

Building upon T3.1, models are additionally required to output `<morph>` JSON (with the same schema as in T1.2). This setup explicitly tests: *whether the morphological evidence documented by the model sufficiently supports its reasoning chain and final diagnosis.*

Consistency check (analysis dimension). Beyond open-ended scoring, we additionally perform automated "morphology↔diagnosis consistency" checks on the core set: For example, contradictions are counted when the model declares critical negative features in the JSON (e.g., *no pigment network*) but cites contradictory evidence in its reasoning.

B.6 Fairness (Task 4.x)

Data and grouping. We reuse the DDI-based 4-way MCQAs and group images according to Fitzpatrick skin type (FST I–V).

Fairness metric. Let Acc_k denote the model accuracy for each group. Fairness is defined as:

$$\text{Fairness} = \frac{\min_k Acc_k}{\max_k Acc_k}.$$

This metric achieves higher values when overall performance is high and performance gaps across skin tone groups are small. In addition to this primary metric, we also report per-group accuracies to avoid misinterpretations where "ratios mask absolute performance differences".

B.7 Gold standard annotation protocol for the 900-case core set

Step 1: Draft generation. Three types of drafts are generated for each image: (i) morphological report, (ii) attribute JSON (Derm7pt/SkinCon schema), and (iii) reasoning chain + final diagnosis.

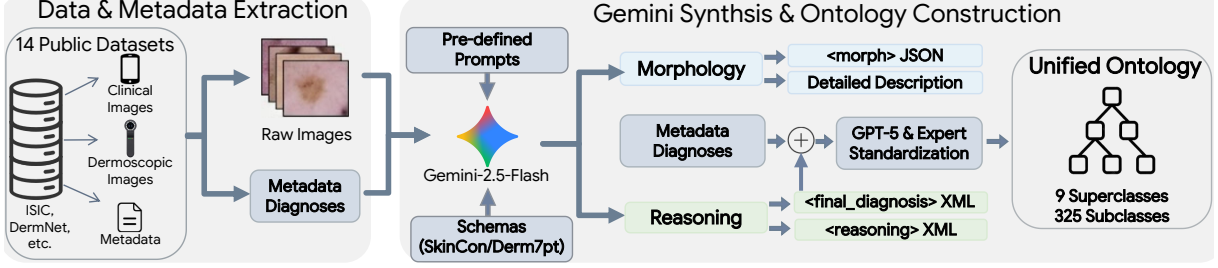


Figure 6: Construction pipeline for DermoInstruct. We aggregate 14 source datasets, apply leakage controls and de-duplication, then generate morphology- and reasoning-grounded instruction pairs using a SOTA multimodal LLM (Gemini-2.5-Flash). The final training subset of DermoInstruct dataset contains 646k high-quality image-instruction pairs.

Step 2: Clinical line-by-line revision. Two dermatologists conduct line-by-line review and revision of the drafts, with focus on correcting: (a) invisible or exaggerated morphological descriptions; (b) JSON field values inconsistent with definitions; (c) reasoning inconsistent with morphological evidence; (d) diagnoses unsupported by the evidence chain.

Step 3: Consistency and format validation. We perform format validation (tag/JSON parseability) and consistency checks (morphology $\leftrightarrow$ reasoning $\leftrightarrow$ diagnosis) for all samples. If conflicts are detected, we return to Step 2 and iterate until all checks pass.

Step 4: Quality spot-checking and documentation. A random subset undergoes dual review by two annotators, with common error patterns documented and revision guidelines updated to ensure annotation consistency and scalability.

B.8 Concept bottleneck tasks

Task motivation. T1.2 and T3.2 enforce the output of standardized morphological concepts (the <morph> JSON), using “interpretable morphological evidence” as a diagnostic intermediate bottleneck. This extends evaluation from merely “whether the answer is correct” to “whether the evidence chain is auditable and self-consistent.”

Output format and ordering constraints. Both tasks require outputting parseable JSON enclosed within <morph>...</morph> tags: (i) T1.2: the <morph> tag is placed before the morphological report; (ii) T3.2: output the <morph> JSON right after the <reasoning> paragraph, and finally the <final_diagnosis>.

C Prompt Templates and Example Outputs for DermoInstruct

C.1 Morphology and Reasoning Supervision

We obtain morphology-centric supervision by querying a SOTA multimodal LLM, Gemini-2.5-Flash (Comanici et al., 2025), with a small set of templates for every image. For each case, the model is asked to (i) describe the lesion in free text, (ii) output a structured set of morphology attributes, and (iii) perform step-by-step diagnostic reasoning that ends in a final diagnosis chosen from a candidate list. This provides a unified image-to-text pipeline whose outputs are reused across DermoInstruct and DermoBench.

We distinguish clinical and dermoscopic images only through the morphology schema. For clinical photographs, prompts align Gemini’s outputs with the 48 SkinCon concepts (Ren et al., 2024), returning a short report plus a JSON object indicating which attributes are present. For dermoscopic photographs, we instead condition on the seven-point checklist (Kawahara et al., 2018) to obtain an analogous JSON over dermoscopic structures and a brief dermoscopy report. In both cases, the model must first commit to morphology before predicting any disease label. We then augment each case with chain-of-thought (CoT) supervision (Wei et al., 2022): given the image, the morphology JSON, and a small candidate set of diagnoses derived from metadata and our ontology (Sec. 2.1.3), Gemini produces a reasoning paragraph and a <final_diagnosis> tag selecting one fine-grained diagnosis.

C.2 Morphology JSON Prompts

C.2.1 Clinical Images (SkinCon)

SkinCon clinical prompt (system + user)

```
PROMPT_DICT = {
  "system_prompt": "You are an expert in dermatology. Your task is to perform a detailed visual analysis of a provided skin lesion image (clinical or dermoscopic). You will be given an image of a skin lesion and a predefined list of 48 standardized clinical concepts from the SkinCon dataset. Your task is to analyze the image, describe it clinically, and then map the observed features to the provided SkinCon concepts. Any features you observe that are not on the list must be categorized separately. Your output must be a single, clean JSON object and nothing else.",
  "user_prompt": "Analyze the provided skin lesion image using the established SkinCon vocabulary. First, perform a detailed, step-by-step visual assessment. Second, generate a single, valid JSON object as your final and ONLY output. Do not include any text, explanations, or markdown formatting outside of the JSON object.\n\n### SkinCon Morphological Concepts List\n\nHere are the 48 standardized concepts you MUST use for classification:\n1. Abscess\n2. Acuminate\n3. Atrophy\n4. Black\n5. Blue\n6. Brown\n7. Hyperpigmentation\n8. Bulla\n9. Burrow\n10. Comedo\n11. Crust\n12. Cyst\n13. Dome-shaped\n14. Erosion\n15. Erythema\n16. Excoriation\n17. Exophytic\n18. Fungating\n19. Exudate\n20. Fissure\n21. Flat topped\n22. Friable\n23. Gray\n24. Induration\n25. Lichenification\n26. Macule\n27. Nodule\n28. Papule\n29. Patch\n30. Pedunculated\n31. Pigmented\n32. Plaque\n33. Poikiloderma\n34. Purple\n35. Purpura/Petechiae\n36. Pustule\n37. Salmon\n38. Scale\n39. Scar\n40. Sclerosis\n41. Telangiectasia\n42. Translucent\n43. Ulcer\n44. Umbilicated\n45. Vesicle\n46. Warty/Papillomatous\n47. Wheal\n48. White (Hypopigmentation)\n49. Xerosis\n50. Yellow\n\n### Required JSON Output Structure\n\nThe JSON object MUST contain exactly three keys:\n1. detailed_description: (String) A comprehensive clinical narrative of the lesion's morphology, including primary lesion type, color, shape, border, surface, and texture.\n2. morphological_features_skincon: (Array of Strings) A list of all observed features that EXACTLY MATCH one or more terms from the 48 SkinCon concepts provided above.\n3. morphological_features_others: (Array of Strings) A list of important
```

```
observed features that are NOT found in the SkinCon list. If all features are covered by the SkinCon list, this array should be empty [].\n\n### Examples for Guidance\n\n**INPUT:** [Image of a Psoriasis Plaque]\n\n**REQUIRED JSON OUTPUT:**\n{\n  \"detailed_description\": \"The image shows a sharply demarcated, erythematous plaque with a raised, indurated surface. The lesion is ovoid and its borders are well-defined. The surface is covered by a thick layer of silvery-white, lamellar scales. The perilesional skin appears unremarkable.\\\",\\n\\\"morphological_features_skincon\": [\\n\\\"Plaque\\\",\\n\\\"Erythema\\\",\\n\\\"Scale\\\"\\n\\\"\\n\\\",\\n\\\"morphological_features_others\": [\\n\\\"well-demarcated\\\",\\n\\\"silvery-white\\\"\\n\\\"\\n\\\"]\\n\\n---\\n\n\n**YOUR TASK**\n\nNow, for the image I have provided, please perform the same analysis and generate the JSON output. Remember, the JSON object is the only thing you should return.\n\n\"}
```

SkinCon example JSON

```
{
  "detailed_description": "The image shows multiple digits (toes) affected by severe onychodystrophy and prominent periungual inflammation. The nail plates are markedly thickened, opaque, and display significant discoloration, predominantly yellow and brownish hues. Many nails exhibit onycholysis, appearing separated from the nail bed, often with underlying subungual hyperkeratosis. The surrounding periungual skin and distal phalanges are diffusely erythematous, swollen, and indurated, indicative of chronic inflammation. Localized areas of scaling and subtle crusting are also observed on the inflamed periungual tissue.",
  "morphological_features_skincon": [
    "Yellow",
    "Brown(Hyperpigmentation)",
    "Erythema",
    "Scale",
    "Induration",
    "Crust"
  ],
  "morphological_features_others": [
    "Onychodystrophy",
    "Onycholysis",
    "Subungual hyperkeratosis",
    "Nail thickening",
    "Opaque nails",
    "Swelling",
    "Periungual inflammation"
  ]
}
```

C.2.2 Dermoscopic Images (Derm7pt)

Derm7pt dermoscopic prompt (system + user)

```
PROMPT_DICT = {
    "system_prompt": "You are an expert in dermatology. Your task is to perform a detailed visual analysis of a provided dermoscopic image. You will analyze the image and classify its features according to the 7-point checklist, assigning the single most fitting morphological label to each of the seven criteria. Your output must be a single, clean JSON object and nothing else.",
    "user_prompt": "Analyze the provided skin lesion image using the established Derm7pt vocabulary. First, perform a detailed, step-by-step visual assessment. Second, for each of the 7 criteria, select the single most appropriate label from the lists provided below. Finally, generate a single, valid JSON object as your final and ONLY output. Do not include any text, explanations, or markdown formatting outside of the JSON object.\n\n### Derm7pt Morphological Concepts and Labels\nYou MUST classify the lesion by selecting exactly one label for each of the 7 criteria:\n1. **pigment_network**: [\n\"absent\", \"typical\", \"atypical\"]\n2. **blue_whitish_veil**: [\n\"absent\", \"present\"]\n3. **vascular_structures**: [\n\"absent\", \"arborizing\", \"comma\", \"hairpin\", \"within regression\", \"wreath\", \"dotted\", \"linear irregular\"]\n4. **pigmentation**: [\n\"absent\", \"diffuse regular\", \"diffuse localized regular\", \"diffuse irregular\", \"localized irregular\"]\n5. **streaks**: [\n\"absent\", \"regular\", \"irregular\"]\n6. **dots_and_globules**: [\n\"absent\", \"regular\", \"irregular\"]\n7. **regression_structures**: [\n\"absent\", \"blue areas\", \"white areas\", \"combinations\"]\n\n### Required JSON Output Structure\nThe JSON object MUST contain exactly three keys:\n1. detailed_description: (String) A comprehensive clinical narrative of the lesion's morphology, including primary lesion type, color, shape, border, surface, and texture, justifying your label choices.\n2. morphological_features_Derm7pt: (Object) An object where each key is one of the 7 Derm7pt criteria and its value is a single (String) label selected from the lists above.\n3. morphological_features_others: (Array of Strings) A list of important observed features that are NOT part of the 7-point checklist classification (e.g., symmetry, specific colors). If none, this array should be empty [].\n\n### Examples for Guidance\n\n**INPUT**\n[Dermoscopic image of a melanoma]\n\n**REQUIRED JSON OUTPUT**:\n{\n  \"detailed_description\": \"Dermoscopy reveals a chaotic and asymmetrical lesion. The pigment network is thickened and irregular, with variable hole sizes and abrupt cut-offs at the periphery, classifying it as 'atypical'. Irregular streaks are visible radiating from the main body. There are multiple blotches of dark brown and black pigment concentrated in one quadrant, consistent with 'localized irregular' pigmentation. Additionally, a peppering of various-sized gray-black dots and globules is present, indicating an 'irregular' pattern. The lesion also features both scar-like white areas and peppercorn-like blue areas, which points to 'combinations' of regression structures. Abnormal linear irregular vessels are noted. A blue-whitish veil is absent.\",\n  \"morphological_features_Derm7pt\": {\n    \"pigment_network\": \"atypical\", \n    \"blue_whitish_veil\": \"absent\", \n    \"vascular_structures\": \"linear irregular\", \n    \"pigmentation\": \"localized irregular\", \n    \"streaks\": \"irregular\", \n    \"dots_and_globules\": \"irregular\", \n    \"regression_structures\": \"combinations\"\n  },\n  \"morphological_features_others\": [\n    \"asymmetry\", \n    \"chaotic appearance\", \n    \"color variegation (dark brown, black, gray-black, white, blue)\"\n  ]\n}\n\n---\n\n**YOUR TASK**\nNow, for the image I have provided, please perform the same analysis and generate the JSON output. Remember, the JSON object is the only thing you should return.\n\n\"}
```

```
.g., symmetry, specific colors). If none, this array should be empty [].\n\n### Examples for Guidance\n\n**INPUT**\n[Dermoscopic image of a melanoma]\n\n**REQUIRED JSON OUTPUT**:\n{\n  \"detailed_description\": \"Dermoscopy reveals a chaotic and asymmetrical lesion. The pigment network is thickened and irregular, with variable hole sizes and abrupt cut-offs at the periphery, classifying it as 'atypical'. Irregular streaks are visible radiating from the main body. There are multiple blotches of dark brown and black pigment concentrated in one quadrant, consistent with 'localized irregular' pigmentation. Additionally, a peppering of various-sized gray-black dots and globules is present, indicating an 'irregular' pattern. The lesion also features both scar-like white areas and peppercorn-like blue areas, which points to 'combinations' of regression structures. Abnormal linear irregular vessels are noted. A blue-whitish veil is absent.\",\n  \"morphological_features_Derm7pt\": {\n    \"pigment_network\": \"atypical\", \n    \"blue_whitish_veil\": \"absent\", \n    \"vascular_structures\": \"linear irregular\", \n    \"pigmentation\": \"localized irregular\", \n    \"streaks\": \"irregular\", \n    \"dots_and_globules\": \"irregular\", \n    \"regression_structures\": \"combinations\"\n  },\n  \"morphological_features_others\": [\n    \"asymmetry\", \n    \"chaotic appearance\", \n    \"color variegation (dark brown, black, gray-black, white, blue)\"\n  ]\n}\n\n---\n\n**YOUR TASK**\nNow, for the image I have provided, please perform the same analysis and generate the JSON output. Remember, the JSON object is the only thing you should return.\n\n\"}
```

C.3 Chain-of-Thought Reasoning Prompt

CoT reasoning prompt (system + user)

```
PROMPT_DICT = {
    "system_prompt": "You are an expert dermatologist AI, acting as a clinical consultant. Your primary task is to analyze a skin lesion image and generate a concise clinical reasoning narrative. You will be provided with potential clinical concepts (which may not be entirely accurate) and a confirmed diagnosis. You must critically evaluate the visual evidence in the image to explain how it supports the diagnosis, adhering to a strict XML format for your output.",
    "user_prompt_template": "Analyze the provided image and its context. Your
```

entire output must be a structured response containing a reasoning block (<reasoning>) and a final diagnosis block (<final_diagnosis>).\n\n### Input Context\n* **Image:**\n* **Potential Clinical Concepts:**\n* { clinical_concepts}\n* **Confirmed Diagnoses:**\n* {diagnoses}\n\n### Your Task\nYour response MUST follow these three rules precisely:\n1. **First,** provide a step-by-step clinical rationale explaining how the visual evidence in the image leads to the confirmed diagnosis. Your explanation should be from the perspective of an expert explaining the case to a colleague. Ground your reasoning in the visual features of the lesion (e.g., shape, color, border, texture, specific structures). Use the 'Potential Clinical Concepts' as a guide, but your primary justification must come from the image itself. Enclose this entire process within <reasoning> and </reasoning> tags.\n2. **Second,** provide the most specific diagnosis from the 'Confirmed Diagnoses' list inside <final_diagnosis> and </final_diagnosis> tags.\n3. **Third,** ensure there is absolutely NO extra text, explanation, or markdown formatting outside of these two required XML tags.\n\n### Example for Guidance\n\n**INPUT CONTEXT:**\n* **Image:** [Dermoscopic image of a melanoma]\n* **Potential Clinical Concepts:** [\"Asymmetry\", \"Irregular Border\", \"Color Variegation (Brown, Black, Blue-Gray)\", \"Atypical Pigment Network\"]\n* **Confirmed Diagnoses:** [\"Malignant\", \"Malignant Melanoma\"]\n\n**REQUIRED OUTPUT:**\n<reasoning>Upon examining the image, the lesion exhibits several hallmark features concerning for malignancy. There is clear asymmetry in its overall shape and the border is poorly defined and irregular, with notches and blurred edges in several areas. I observe significant color variegation, with multiple shades of brown and black, as well as a focal blue-gray area, which is a strong indicator of pigment regression or deep melanin. These observations align with the classic clinical signs for melanoma. The combination of these visual findings provides a strong basis for diagnosing this lesion as a malignant melanoma, differentiating it from a benign nevus.</reasoning><final_diagnosis>Malignant Melanoma</final_diagnosis>\n\n---\n\n### YOUR TASK\nNow, for the image, concepts, and diagnoses I have provided, generate the response in the required format."

}

CoT example XML

```
<reasoning>Upon visual inspection, the image displays an erythematous, ill-defined plaque with an irregular shape on the skin. Centrally, there are multiple confluent erosions and ulcerations, appearing moist and suggestive of serous exudate. A yellowish-brown crust is also visible within this central eroded area, indicating dried serous fluid or possibly a secondary bacterial component. The presence of acute erythema, clustered erosions, exudate, and crusting is highly characteristic of an acute viral infection, such as Herpes Simplex Virus. This morphology strongly supports the diagnosis as fitting within the category of "other Viral Infections," as distinct from typical warts or molluscum contagiosum which present differently.</reasoning><final_diagnosis>Warts Molluscum and other Viral Infections</final_diagnosis>
```

C.4 Diagnosis VQA prompt templates

Using the ontology described above, we synthesize diagnosis VQA items in two forms. First, for flat four-way MCQA questions, we sample one ground-truth diagnosis and three ontology-consistent distractors (typically siblings or closely related conditions), and render them as options A–D. The question stem is drawn at random from a small pool of interchangeable prompts that ask the model to choose the most likely diagnosis. This yields diverse yet semantically equivalent formulations while keeping the underlying label space fixed.

Second, for hierarchical diagnosis VQA, we traverse the ontology level by level. At each step, we present the image and a set of candidate categories, and instantiate one of several templated prompts for (i) selecting a top-level superclass, (ii) refining the choice within its subcategories, and (iii) choosing a final leaf diagnosis. Additional declarative prompts are used to convert the completed path into a natural-language statement of the final diagnosis, and a small set of “human correction” prompts supports expert editing when the automatically proposed path is incorrect.

Together, these instruction types give dense supervision over both *what* diagnosis to output and *how* to traverse and correct a hierarchical diagnostic reasoning process.

PROMPTS for 4-way diagnosis MCQA

```
PROMPTS = [  
    "Observe this skin image. Which of the  
    following diagnoses is the most  
    likely?",  
    "Based on the skin lesion shown in this  
    image, please select the most  
    accurate diagnosis from the options  
    below.",  
    "Which of the following diagnoses best  
    matches the skin condition shown in  
    this image?",  
    "Considering the clinical presentation  
    of the skin lesion in the image,  
    which of the following is the most  
    likely diagnosis?"  
]
```

TOP_LEVEL_PROMPTS_GEN

```
TOP_LEVEL_PROMPTS_GEN = [  
    "Based on the clinical image, identify  
    the most fitting major  
    dermatological category from the  
    following list: {options_list}.",  
    "Observe the skin lesion. Which of these  
    high-level classifications best  
    describes it? Here are the  
    possibilities: {options_list}.",  
    "Please provide a broad categorization  
    for the skin condition shown. Your  
    answer should be one of the  
    following: {options_list}."  
]
```

SUB_LEVEL_PROMPTS_GEN

```
SUB_LEVEL_PROMPTS_GEN = [  
    "Correct, the condition is a form of '{  
    parent_category}'. Now, specify the  
    sub-category from this list: {  
    options_list}.",  
    "Proceeding from '{parent_category}',  
    which of the following groups does  
    this lesion belong to? {options_list  
    }.",  
    "Understood. Let's refine the diagnosis  
    within '{parent_category}'. Please  
    choose the most accurate description  
    from the following: {options_list  
    }."  
]
```

FINAL_LEVEL_PROMPTS_GEN

```
FINAL_LEVEL_PROMPTS_GEN = [  
    "We've classified this under '{  
    parent_category}'. Now, provide the  
    definitive diagnosis from the  
    choices available: {options_list}.",  
    "Excellent. To finalize, please state  
    the specific diagnosis for '{
```

```
parent_category} ', which should be  
one of the following: {options_list  
}."",  
    "Perfect. Based on our hierarchical  
    classification ending with '{  
    parent_category}', please identify  
    the definitive diagnosis from this  
    list: {options_list}."  
]
```

DECLARATIVE_PROMPTS

```
DECLARATIVE_PROMPTS = [  
    "Following the diagnostic path to '{  
    parent_category}', the evidence  
    points to a single definitive  
    diagnosis, which is {final_diagnosis  
    }.",  
    "Correct. The reasoning has led us to '{  
    parent_category}', which contains  
    only one specific condition.  
    Therefore, the diagnosis must be {  
    final_diagnosis}.",  
    "Excellent. Since '{parent_category}' is  
    the most specific category and it  
    corresponds to a single diagnosis,  
    we can conclude the condition is {  
    final_diagnosis}."  
]
```

HUMAN_CORRECTION_PROMPTS

```
HUMAN_CORRECTION_PROMPTS = [  
    "That's not quite right. While '{  
    wrong_choice}' is a possibility, the  
    visual evidence points more  
    strongly to '{correct_choice}'. Let'  
    s proceed with the correct category  
    .",  
    "Actually, that's incorrect. A closer  
    look reveals features more  
    consistent with '{correct_choice}'.  
    Please correct the path.",  
    "Incorrect. The diagnosis should be '{  
    correct_choice}', not '{wrong_choice  
    }'. Let's continue from the right  
    category.",  
    "I disagree. '{correct_choice}' is the  
    more accurate classification here.  
    Let's use that one instead."  
]
```

D LLM-as-a-Judge Prompts

We use a text-only LLM-as-a-Judge protocol: the judge *does not see the image* and evaluates by comparing the **REFERENCE** text versus the **CANDIDATE** text under a strict dermatology morphology rubric. All tasks output a scalar `final_overall` in $[0, 100]$ and we report `mean_final_overall` in the main paper.

D.1 Task 1.1 (Morph Description)

Task 1.1 – SYSTEM PROMPT

You are a strict, no-nonsense clinical dermatology evaluator.
You DO NOT see the image; evaluate ONLY by comparing the REFERENCE vs the CANDIDATE text.
Use dermatology morphology standards. Avoid rewarding verbosity; penalize contradictions and invented findings.
Focus on: anatomical site, number/arrangement, primary lesion types, color, shape, borders, surface features, size/extent, distribution/pattern, and special/contextual features (e.g., pen markings, dermoscopic 7-point structures if applicable).
Return STRICT JSON only.

Task 1.1 – USER PROMPT TEMPLATE

[Task Prompt]
\{task_prompt\}

[REFERENCE]
\{reference\}

[CANDIDATE]
\{candidate\}

Evaluate as follows:

- 1) Decompose REFERENCE into ≤ 25 atomic CLAIMS.
- 2) For each CLAIM, label wrt CANDIDATE: Supported, PartiallySupported, Contradicted, Missing, or Vague.
- 3) Identify any EXTRA INCORRECT statements in CANDIDATE.
- 4) Score:
$$\text{recall_like} = (\text{Supported} + 0.5 * \text{PartiallySupported}) / \max(1, \text{total_ref_claims})$$
$$\text{precision_penalty} = \min(1.0, (\text{Contradicted} + \text{ExtraIncorrect}) / \max(1, \text{total_ref_claims}))$$
$$\text{overall} [0-100] = \text{round}(100 * \max(0, \text{recall_like} - 0.5 * \text{precision_penalty}), 1)$$

Provide rubric sub-scores (accuracy, completeness, consistency) in [0,1].

JSON ONLY. Schema:

```
\{
  "claims": [{"text": "...", "label": "Supported|PartiallySupported|Contradicted|Missing|Vague"}],
  "counts": {"supported": 0, "partial": 0, "contradicted": 0, "missing": 0, "vague": 0, "extra\_incorrect": 0, "total\_ref\_claims": 0},
  "rubric": {"accuracy": 0.0, "completeness": 0.0, "consistency": 0.0},
  "overall": 0.0,
}
```

```
"short\_feedback": "<=40 words concise justification"
\}
```

D.2 Task 1.2 (Morph Content + Narrative)

Task 1.2 – SYSTEM PROMPT

You are a strict dermatology evaluator for Task 1.2 (morph content + narrative).
You DO NOT see the image. Focus on CONTENT, not formatting.
Both REFERENCE and CANDIDATE may or may not wrap the morph JSON in <morph> tags.
Do NOT penalize missing tags, extra whitespace, or minor ordering/format differences.
If a JSON block is present anywhere, treat the FIRST JSON object as the morph content.
If no JSON is present, infer the morph feature set from the surrounding text.
Schemas you may encounter:
- SkinCon: \{"morphological_features_skincon": [<feature strings>]\}
- Derm7pt: \{"morphological_features_Derm7pt": \{pigment_network, blue_whitish_veil, vascular_structures, pigmentation, streaks, dots_and_globules, regression_structures\}\}
For the narrative comparison, use dermatology morphology standards (site, number/arrangement, primary lesion types, color, shape, borders, surface features, size/extent, distribution/pattern, special/context).
Also check CROSS-CONSISTENCY between the CANDIDATE morph content and CANDIDATE narrative.
Return STRICT JSON only.

Task 1.2 – USER PROMPT TEMPLATE

You will be given REFERENCE and CANDIDATE texts.
Each may contain a morph JSON (SkinCon or Derm7pt) with or without <morph> tags, possibly followed by a narrative paragraph. Do NOT penalize formatting.

Rules:

- If a JSON object appears anywhere, treat the FIRST JSON object as the morph content.
- If no JSON is found, infer the morph feature set from the surrounding text (best-effort).
- Use synonyms tolerance for semantic matching.

[Task Prompt]
\{task_prompt\}

[REFERENCE]
\{reference\}

```
[CANDIDATE]
\{candidate\}
```

Your tasks:

- 1) MORPH SEMANTICS (content-first): Compare CANDIDATE-morph vs REFERENCE-morph semantically (synonyms allowed). Count supported/missing/contradicted/extra and give a semantic score in [0,1]. If CANDIDATE has no explicit JSON, infer its morph set from the candidate text.
- 2) TEXT (NARRATIVE): Compare REFERENCE-narrative vs CANDIDATE-narrative using morphology standards. Extract ≤ 25 atomic claims from the REFERENCE-narrative; for each, label CANDIDATE as Supported/PartiallySupported/Contradicted/Missing/Vague. Provide rubric sub-scores (accuracy, completeness, consistency) in [0,1] and overall [0,100] using:
$$\text{recall_like} = (\text{Supported} + 0.5 * \text{PartiallySupported}) / \max(1, \text{total_ref_claims})$$
$$\text{precision_penalty} = \min(1.0, (\text{Contradicted} + \text{ExtraIncorrect}) / \max(1, \text{total_ref_claims}))$$
$$\text{overall} = \text{round}(100 * \max(0, \text{recall_like} - 0.5 * \text{precision_penalty}), 1)$$
- 3) CROSS-CONSISTENCY: Judge if the CANDIDATE narrative contradicts the CANDIDATE morph content. Output a penalty in [0,1] (0=no issue, 1=severe) and short notes.

Output STRICT JSON:

```
\{
  "morph\_semantic": \{
    "schema": "SkinCon" | "Derm7pt" | "Unknown",
    "supported": 0, "missing": 0, "contradicted": 0, "extra": 0,
    "score\_semantic": 0.0,
    "notes": "<=60 words"
  },
  "text\_judge": \{
    "claims": [\{"text": "...", "label": "Supported|PartiallySupported|Contradicted|Missing|Vague"\}],
    "counts": \{"supported":0, "partial":0, "contradicted":0, "missing":0, "vague":0, "extra\_incorrect":0, "total\_ref\_claims":0\},
    "rubric": \{"accuracy":0.0, "completeness":0.0, "consistency":0.0\},
    "overall": 0.0,
    "short\_feedback": "<=40 words"
  },
  "cross\_consistency": \{"penalty": 0.0, "notes": "<=40 words"\}
}
```

D.3 Task 3.1 (Reasoning + Final Diagnosis)

Task 3.1 – SYSTEM PROMPT

You are a strict dermatology evaluator for Task 3 (reasoning + final diagnosis). You DO NOT see the image; evaluate ONLY the textual content. Ignore formatting and tags.

Goal: robustly extract (A) the candidate's reasoning and (B) the candidate's final diagnosis, then score (1) REASONING ALIGNMENT vs the GT reasoning and (2) DIAGNOSIS SIMILARITY vs the GT final diagnosis.

Penalize contradictions and hallucinated findings. Do not reward verbosity. Return STRICT JSON only.

Task 3.1 – USER PROMPT TEMPLATE

```
[Task Prompt]
\{task\_prompt\}
```

```
[GROUND\_TRUTH\_RAW]
\{reference\}
```

```
[CANDIDATE\_RAW]
\{candidate\}
```

Evaluate with these steps (format-agnostic; focus on content):

- A) Extraction (be robust even if the candidate is unstructured):
 - From GROUND_TRUTH_RAW, extract:
gt_reasoning: inside <reasoning>...</reasoning> if present; else best-effort summary.
gt_final_dx: inside <final_diagnosis>...</final_diagnosis> if present; else best-effort label.
 - From CANDIDATE_RAW, extract:
cand_reasoning: the explanation/rationale (anywhere).
cand_final_dx: the single most likely final diagnosis term/phrase.
- B) Reasoning Alignment:
 - Decompose gt_reasoning into ≤ 25 atomic claims.
 - For each claim, label wrt cand_reasoning: Supported | PartiallySupported | Contradicted | Missing | Vague.
 - Compute reasoning_score [0-100] using the same recall/penalty formula.
- C) Diagnosis Similarity (graded, not binary):
 - Decide relation: Exact | Synonym | Parent | Child | Sibling | CloseDifferential | SameSuperfamily | UnrelatedPlausible | WrongSystem | Nonsense/NoAnswer.
 - Map to similarity in [0,1] and compute diagnosis_score [0-100].

D) Overall:
- overall [0-100] = round(0.5 * reasoning_score + 0.5 * diagnosis_score, 1)

STRICT JSON ONLY (use the specified schema in the paper).

D.4 Task 3.2 (Morph-grounded Reasoning)

Task 3.2 – SYSTEM PROMPT

You are a strict dermatology evaluator for Task 3.2 (reasoning + morph JSON + final diagnosis).
You DO NOT see the image. Focus on CONTENT, not formatting.
Both REFERENCE and CANDIDATE may or may not wrap the morph JSON in <morph> tags.
Do NOT penalize missing tags, extra whitespace, or ordering differences.
If a JSON object appears anywhere, treat the FIRST JSON object as the morph content.
If no JSON is present, infer the morph feature set from the surrounding text.
SCHEMA SELECTION RULE: Detect the schema used by REFERENCE. Compare and output using the SAME schema.

Task 3.2 – USER PROMPT TEMPLATE

You will be given REFERENCE and CANDIDATE texts containing three conceptual parts: <reasoning>, <morph> JSON, and <final_diagnosis>.
Be format-agnostic; extract content even when tags are missing or order differs.

Allowed schemas:

- Derm7pt (object with EXACT keys): pigment_network, blue_whitish_veil, vascular_structures, pigmentation, streaks, dots_and_globules, regression_structures
- SkinCon (array of strings only): \{"morphological_features_skincon": [...]\} from a CLOSED set.

SCHEMA SELECTION:

- Detect the schema used by REFERENCE (Derm7pt vs SkinCon). Use that schema for extraction/normalization and comparison. Do NOT switch schemas.

[Task Prompt]
\{task_prompt\}

[REFERENCE]
\{reference\}

[CANDIDATE]
\{candidate\}

Tasks:

- A) EXTRACTION: reasoning, morph (normalized to REFERENCE schema), final_dx for both sides.
- B) REASONING ALIGNMENT: compute reasoning_score [0-100].
- C) MORPH SEMANTICS: score_semantic in [0,1].
- D) DIAGNOSIS SIMILARITY: diagnosis_score [0-100].
- E) CROSS-CONSISTENCY: penalty in [0,1] if candidate reasoning contradicts candidate morph JSON.

STRICT JSON ONLY (use the specified schema in the paper).

D.5 Judge Reliability and Human Sanity Check

D.5.1 Judge sensitivity on the 900-case core set

Table 7 reports mean_final_overall on the 900-case core set when swapping the judge between Gemini-2.5-Pro (main paper default) and GPT-5. This comparison is intended as a robustness check for evaluator choice rather than a replacement of the main evaluation protocol.

Candidate model	Judge	T1.1	T1.2	T3.1	T3.2
Qwen3-VL-8B	Gemini-2.5-Pro	33.18	46.05	47.53	53.43
Qwen3-VL-8B	GPT-5	37.73	43.92	51.08	59.81
GPT-4o-mini	Gemini-2.5-Pro	34.55	51.80	42.83	51.65
GPT-4o-mini	GPT-5	31.32	47.82	45.28	49.17

Table 7: Judge sensitivity on the 900-case core set (reported as mean_final_overall in [0, 100]).

D.5.2 Aggregate-level inter-judge agreement metrics

Using the 8 paired items in Table 7 (2 candidate models × 4 tasks), we compute rank/absolute agreement metrics between GPT-5 and Gemini-2.5-Pro judge scores. Results indicate strong agreement at the level of model-task means.

Metric	Value
Pearson r	0.883
Spearman ρ	0.857
Mean difference (GPT-5 – Gemini)	+0.65
Mean absolute difference (MAE)	3.60

Table 8: Inter-judge agreement between GPT-5 and Gemini-2.5-Pro computed over the 8 paired model-task means in Table 7.

D.6 Human sanity check (20 cases)

We further sample 20 cases from Qwen3-VL-8B + Gemini-2.5-Pro and ask clinicians to rate whether

the judge scoring and feedback are reasonable on a 0–5 scale (higher is more reasonable). Figure 2c summarizes the reasonableness ratings.

E Training Details

E.1 Hyperparameters

Backbone and precision. We initialize from Qwen3-VL-8B-Instruct, train with Deepspeed ZeRO-2, and use BF16 with TF32 enabled. FlashAttention-2 is used unless stated otherwise. Gradient checkpointing is enabled in both stages.

Stage 1: Supervised fine-tuning (SFT). We perform one epoch of multi-task SFT on the merged instruction data. We enable LoRA adapters with rank $r=64$, $\alpha=64$, dropout 0.05, and exclude `lm_head` and `embed_tokens` from LoRA injection. We freeze the language model backbone (`freeze_llm=True`), while keeping the vision tower and merger trainable (`freeze_vision_tower=False`, `freeze_merger=False`). We set per-device batch size to 8 on 8 GPUs with gradient accumulation steps 2 (global batch size 128). We train with learning rate $1e-4$, and optionally use module-specific learning rates for the vision tower ($2e-6$) and the merger ($1e-5$). Weight decay is 0.1, warmup ratio is 0.03, and we use a cosine scheduler. Images are resized by pixel constraints with `image_min_pixels = 256 · 322` and `image_max_pixels = 1280 · 322`. Unless otherwise specified, we use the training framework’s default AdamW-type optimizer settings.

Stage 2: GRPO with MAVIC reward. We further optimize the SFT checkpoint with GRPO using group size $K=\text{num_generations} = 8$. We train for one epoch with per-device batch size 32 and gradient accumulation steps 3. We sample completions with temperature 1.0, top- p 1.0, and top- k 50, using maximum prompt length 4096 and maximum completion length 640. We set learning rate to $1e-6$, weight decay to 0.1, warmup ratio to 0.03, and cosine scheduler. We use $\text{beta}=0.1$ for GRPO’s KL regularization. In this stage, we freeze the vision tower, language model, and merger, and train only LoRA adapters (LoRA rank 16, $\alpha = 32$, dropout 0.05, excluding `lm_head` and `embed_tokens`). Images are constrained by `image_min_pixels = 256 · 282` and `image_max_pixels = 1280 · 282`.

Hyperparameter	SFT	GRPO
GPUs	8	8
Epochs	1	1
Per-device batch	8	32
Grad. accumulation	2	3
Global batch	128	768
LoRA rank / α	64 / 64	16 / 32
LoRA dropout	0.05	0.05
Backbone frozen?	LLM frozen	LLM/Vision/Merger frozen
LR	$1e-4$	$1e-6$
Vision LR / Merger LR	$2e-6$ / $1e-5$	–
Weight decay	0.1	0.1
Warmup / Scheduler	0.03 / cosine	0.03 / cosine
Group size K	–	8
Sampling	–	$T=1.0$, top- $p=1.0$, top- $k=50$
Max prompt / completion	–	4096 / 640
KL coef. (beta)	–	0.1

Table 9: Key hyperparameters for SFT and RL training.

E.2 MAVIC Implementation Details

Morphology representation (tokens). Each completion must contain a structured morphology field encoded as JSON under a `<morph>` tag. For dermoscopic images, we use Derm7pt-style attributes (Kawahara et al., 2018); for clinical images, we use SkinCon-style attributes (Ren et al., 2024). We binarize morphology into a vector $\mathbf{m} \in \{0,1\}^F$, where each dimension f corresponds to an attribute indicator. For Derm7pt, we expand categorical states into attribute-state indicators (e.g., `streaks_irregular`); for SkinCon, each label is an indicator.

PMI-based weights (precomputed lookup). Because each training sample has a known leaf diagnosis y , we precompute diagnosis-conditioned weights $w_f(y)$ *once* before RL training. We estimate PMI with log and $\epsilon = 10^{-5}$ smoothing and keep negative values:

$$\text{PMI}(m_f; y) = \log \frac{\hat{p}(m_f=1, y) + \epsilon}{\hat{p}(m_f=1)\hat{p}(y) + \epsilon}. \quad (5)$$

We then normalize per diagnosis with a softmax over features:

$$w_f(y) = \frac{\exp(\text{PMI}(m_f; y))}{\sum_{f'} \exp(\text{PMI}(m_{f'}; y))}. \quad (6)$$

During RL, $w_f(y)$ is obtained by table lookup.

Morphology similarity S_{morph} . Let P and G be the predicted and ground-truth sets of active morphology indicators. We compute a PMI-weighted

Tversky score with $\alpha = 0.7, \beta = 0.3$:

$$\begin{aligned} \text{TP} &= \sum_f w_f \mathbf{1}[\hat{m}_f = 1 \wedge m_f = 1], \\ \text{FP} &= \sum_f w_f \mathbf{1}[\hat{m}_f = 1 \wedge m_f = 0], \\ \text{FN} &= \sum_f w_f \mathbf{1}[\hat{m}_f = 0 \wedge m_f = 1]. \end{aligned} \quad (7)$$

$$S_{\text{morph}}(\hat{\mathbf{m}}, \mathbf{m}) = \frac{\text{TP}}{\text{TP} + \alpha \text{FP} + \beta \text{FN}}. \quad (8)$$

Hierarchy similarity S_{hier} . We map a diagnosis to its taxonomy path (ancestors) and append the leaf label to the end of the path. We compute Wu–Palmer similarity:

$$S_{\text{hier}} = \frac{2 \cdot \text{depth}(\text{LCA}(\text{path}_{\text{pred}}, \text{path}_{\text{gt}}))}{|\text{path}_{\text{pred}}| + |\text{path}_{\text{gt}}|}. \quad (9)$$

When parsing model outputs, we canonicalize strings and use alias/fuzzy matching (threshold 0.8) to map predictions to taxonomy leaves.

Soft gate. Within each GRPO sampling group (size K), we set μ as the median S_{hier} and apply the sigmoid gate with $k = 10$.

Format term R_{fmt} . $R_{\text{fmt}} \in \{0, 1\}$ indicates whether the completion satisfies required tag structure and JSON validity: (i) presence of required tags (e.g., `<morph>` and, for reasoning tasks, `<final_diagnosis>`); (ii) parseable JSON under `<morph>`; (iii) exactly one valid schema (Derm7pt or SkinCon); (iv) schema matches image modality; and (v) tag ordering constraints when applicable. Invalid outputs receive $R_{\text{fmt}} = 0$.

Hyperparameters. We use $\lambda_{\text{hier}} = \lambda_{\text{morph}} = 1$, $\alpha = 0.7$, $\beta = 0.3$, $\epsilon = 10^{-5}$, fuzzy threshold 0.8, and gate slope $k = 10$.

F Ablation Study

F.1 Impact of MAVIC Reward Components

As shown in Table 4, using standard reinforcement learning rewards alone (acc+fmt) actually degrades performance on T3.2 (59.88). Incorporating morphological similarity reward S_{morph} and hierarchical diagnosis reward S_{hier} steadily improves scores to 65.48. Crucially, the combination of S_{morph} with the logical gating mechanism $g(S_{\text{hier}})$ effectively prevents models from bypassing pathological features to make uninformed diagnostic guesses.

K	Task2.4 (OOD) $\uparrow$	Task4 (Fair.) $\uparrow$
2	65.82	93.81
4	66.27	93.76
8	66.48	93.88

Table 10: Sensitivity to the number of prompt variants K .

F.2 Ablation of Confidence–Consistency Components

Setup. We evaluate test-time adaptation (TTA) under the same deterministic decoding setting as the main paper (temperature = 0). The only source of diversity is prompt paraphrasing: we use K prompt variants per example (including the original prompt), and aggregate MCQA option probabilities derived from the first-step logits.

Baselines. We compare against standard, simpler ensemble decoding variants: (i) **Single** ($K=1$), no TTA; (ii) **Vote**, majority vote over predicted option letters across prompts; (iii) **MeanProb**, unweighted averaging of option probability vectors $\mathbf{p}_r$; (iv) **ConfOnly**, weights based on confidence margin only ($\beta=0$); (v) **ConsOnly**, weights based on consistency only (drop $\tilde{C}_r$ term); (vi) **CC (Ours)**, full confidence–consistency weighting.

Sensitivity to K and hyperparameters. We further vary the number of prompt variants K and the confidence exponent α / consistency weight β .

Takeaway. Across datasets, the gains of CC aggregation cannot be explained solely by using more prompts (K), and persist after controlling for simpler voting/averaging baselines, supporting the claim that *confidence* and *consistency* provide complementary signals for robust MCQA aggregation.

G Theoretical Analysis

We provide a probabilistic model explaining why our CCT can suppress outlier rollouts and remain close to an underlying “ideal” token distribution.

Setup. Fix a decoding step t . For notational simplicity, we omit the superscript and write $p_r \in \Delta^{V-1}$ for the token distribution of the r -th rollout at this step, where Δ^{V-1} is the probability simplex in $\mathbb{R}^V$. For any $p \in \Delta^{V-1}$ we have

$$\|p\|_2 \leq 1, \quad (10)$$

and hence for any $p, p^* \in \Delta^{V-1}$,

$$\|p - p^*\|_2^2 \leq 2. \quad (11)$$

At this time step, our method forms a weighted ensemble

$$q = \sum_{r=1}^K w_r p_r, \quad w_r = \frac{\exp(\lambda C_r - \beta D_r)}{\sum_{j=1}^K \exp(\lambda C_j - \beta D_j)}. \quad (12)$$

where

- $C_r \in [0, 1]$ is a margin-based confidence score, derived from the top-1 vs. top-2 probability gap of p_r ;
- $D_r = \frac{1}{2} \|p_r - \bar{p}\|_2^2$ is the squared ℓ_2 -distance to the empirical barycenter $\bar{p} := \frac{1}{K} \sum_{j=1}^K p_j$;
- $\lambda \geq 0$ controls the strength of the confidence term, and $\beta > 0$ controls how aggressively we downweight outliers.

Intuitively, D_r penalizes rollouts that deviate from the main cluster, while C_r slightly favors locally confident rollouts among those that are consistent.

We now formalize this intuition via a contamination model.

G.1 Huber Contamination on the Simplex

We assume that the rollouts at a fixed decoding step are i.i.d. samples from a mixture of a “clean” (good) component and a contaminated (bad) component.

Assumption 1 (Huber contamination on the simplex). *There exists an unknown target distribution $p^* \in \Delta^{V-1}$ such that each rollout distribution p_r is drawn i.i.d. from*

$$p_r \sim (1 - \varepsilon) \mathcal{D}_G + \varepsilon \mathcal{D}_B, \quad (13)$$

where $r = 1, \dots, K$, $0 \leq \varepsilon < \frac{1}{2}$, $\mathcal{D}_G$ and $\mathcal{D}_B$ denote the clean and contaminated components, respectively.

We assume the following moment and separation conditions:

$$\mathbb{E}_{p \sim \mathcal{D}_G} [\|p - p^*\|_2^2] \leq \sigma^2, \quad (14)$$

$$\mathbb{E}_{p \sim \mathcal{D}_B} [\|p - p^*\|_2^2] \geq \sigma^2 + \Delta^2, \quad (15)$$

for some $\sigma^2 > 0$ and $\Delta^2 > 0$. Let $\mu_G := \mathbb{E}_{\mathcal{D}_G}[p]$ and $\mu_B := \mathbb{E}_{\mathcal{D}_B}[p]$ be the means of the clean and contaminated components, respectively. We further assume a signal-to-noise condition:

$$\varepsilon \|\mu_B - \mu_G\|_2 \leq c_0 \Delta \quad \text{for some } c_0 < \frac{1}{2} \quad (16)$$

Finally, we assume that the clean noise level σ is sufficiently small relative to the separation Δ

(and the contamination rate ε) so that there exists a parameter $\alpha \in (0, 1)$ satisfying simultaneously:

$$R_G(\alpha) := \frac{\sigma}{\sqrt{\alpha}} < R_B := \sqrt{\sigma^2 + \frac{\Delta^2}{2}}, \quad (17)$$

$$(1 - \varepsilon)(1 - \alpha) > \frac{1}{2}, \quad (18)$$

$$\sigma + c_0 \Delta \leq \eta(R_B - R_G) \quad (19)$$

for some $\eta \in (0, \frac{1}{2})$. This mild requirement is automatically satisfied whenever the clean cluster is sufficiently concentrated (small σ) compared to the separation Δ and the contamination rate ε is moderate.

Assumption 1 is a Huber contamination model adapted to the probability simplex. Conditions (14)–(15) ensure that the clean component concentrates around p^* , while the contaminated component is, on average, farther away. The signal-to-noise condition (16) ensures that the mixture mean is not dominated by the contaminated component. Conditions (17)–(18) guarantee that we can choose a single parameter α that yields both geometric separation and a strict majority of “good” rollouts.

Because $p_r \in \Delta^{V-1}$, all random variables are uniformly bounded by (10), and standard concentration inequalities (Hoeffding, Chernoff, and their vector-valued variants) apply directly.

G.2 High-Probability Geometric Separation

We now show that, under Assumption 1, the empirical sample $\{p_r\}_{r=1}^K$ exhibits a geometric “good-cluster / bad-cluster” separation with high probability. This is precisely the structure used in deterministic analyses of outlier suppression.

Lemma 1 (High-probability geometric separation). *Suppose Assumption 1 holds and the rollouts $p_1, \dots, p_K$ are drawn i.i.d. from the mixture (13). Fix any $\delta \in (0, 1)$ and let $\alpha \in (0, 1)$ be chosen so that (17) and (18) hold. Define*

$$\begin{aligned} \varepsilon_{\text{eff}} &:= R_G(\alpha) = \frac{\sigma}{\sqrt{\alpha}}, \\ \Delta_{\text{eff}} &:= R_B = \sqrt{\sigma^2 + \frac{\Delta^2}{2}}. \end{aligned} \quad (20)$$

Then there exist constants $\rho_{\text{eff}} \in (\frac{1}{2}, 1)$, $\eta \in (0, \frac{1}{2})$ and a sample size threshold $K_0 = K_0(\sigma, \Delta, \varepsilon, \alpha, \delta)$ such that the following holds.

If $K \geq K_0$, then with probability at least $1 - \delta$ over the draw of $\{p_r\}_{r=1}^K$, there exist index sets $G_{\text{eff}}, B_{\text{eff}} \subseteq \{1, \dots, K\}$ with $G_{\text{eff}} \cap B_{\text{eff}} = \emptyset$ and $G_{\text{eff}} \cup B_{\text{eff}} \neq \emptyset$ such that:

1. (Effective good cluster)

$$\begin{aligned} \|p_g - p^*\|_2 &\leq \varepsilon_{\text{eff}}, \forall g \in G_{\text{eff}}, \\ |G_{\text{eff}}| &\geq \rho_{\text{eff}} K. \end{aligned} \quad (21)$$

where $\rho_{\text{eff}} > \frac{1}{2}$.

2. (Effective bad cluster is farther)

$$\begin{aligned} \|p_b - p^*\|_2 &\geq \Delta_{\text{eff}}, \forall b \in B_{\text{eff}}, \\ \Delta_{\text{eff}} &> \varepsilon_{\text{eff}}. \end{aligned} \quad (22)$$

3. (Barycenter remains in the attraction basin)

Let $\bar{p} := \frac{1}{K} \sum_{r=1}^K p_r$ be the empirical barycenter. Then

$$\|\bar{p} - p^*\|_2 \leq \eta (\Delta_{\text{eff}} - \varepsilon_{\text{eff}}). \quad (23)$$

Proof. We proceed in three steps.

Step 1: Effective good cluster. Consider the random variable

$$X_G(p) := \|p - p^*\|_2^2,$$

for $p \sim \mathcal{D}_G$. By (14), $\mathbb{E}_{\mathcal{D}_G}[X_G] \leq \sigma^2$, and by (11), $0 \leq X_G(p) \leq 2$ a.s.

By Markov's inequality, for the fixed $\alpha \in (0, 1)$ (chosen in the assumption),

$$\Pr_{p \sim \mathcal{D}_G} (X_G(p) > \frac{\sigma^2}{\alpha}) \leq \alpha. \quad (24)$$

Equivalently,

$$\begin{aligned} \Pr_{p \sim \mathcal{D}_G} \left(\|p - p^*\|_2 \leq \frac{\sigma}{\sqrt{\alpha}} \right) &= \Pr_{p \sim \mathcal{D}_G} \left(X_G(p) \leq \frac{\sigma^2}{\alpha} \right), \\ &\geq 1 - \alpha. \end{aligned} \quad (25)$$

Recall that we define

$$R_G(\alpha) := \frac{\sigma}{\sqrt{\alpha}}, \varepsilon_{\text{eff}} := R_G(\alpha).$$

Now consider the mixture $\mathcal{D}$ in (13). The probability that p is drawn from $\mathcal{D}_G$ and satisfies $\|p - p^*\|_2 \leq R_G(\alpha)$ is at least

$$\begin{aligned} \Pr_{p \sim \mathcal{D}} \left(p \sim \mathcal{D}_G, \|p - p^*\|_2 \leq R_G(\alpha) \right) \\ \geq (1 - \varepsilon)(1 - \alpha). \end{aligned} \quad (26)$$

where we used independence between the mixture component choice and the conditional distribution.

For each $r \in \{1, \dots, K\}$, define the indicator

$$I_r := \mathbf{1}\{p_r \sim \mathcal{D}_G \text{ and } \|p_r - p^*\|_2 \leq R_G(\alpha)\}.$$

Then $(I_r)_{r=1}^K$ are i.i.d. Bernoulli random variables with

$$\mathbb{E}[I_r] = \Pr_{p_r \sim \mathcal{D}} (I_r = 1) \geq (1 - \varepsilon)(1 - \alpha). \quad (27)$$

By Hoeffding's inequality, for any $\tau > 0$,

$$\begin{aligned} \Pr \left(\frac{1}{K} \sum_{r=1}^K I_r \leq (1 - \varepsilon)(1 - \alpha) - \tau \right) \\ \leq \exp(-2K\tau^2). \end{aligned} \quad (28)$$

Since by Assumption (18), $(1 - \varepsilon)(1 - \alpha) > \frac{1}{2}$, we can choose $\tau > 0$ such that

$$(1 - \varepsilon)(1 - \alpha) - \tau > \frac{1}{2}.$$

Fix such a τ , and define the event

$$\mathcal{E}_G := \left\{ \frac{1}{K} \sum_{r=1}^K I_r > (1 - \varepsilon)(1 - \alpha) - \tau \right\}.$$

Given a target failure probability $\delta \in (0, 1)$, choose K large enough such that

$$\exp(-2K\tau^2) \leq \frac{\delta}{3}.$$

Then $\Pr(\mathcal{E}_G) \geq 1 - \delta/3$, and on $\mathcal{E}_G$,

$$\sum_{r=1}^K I_r > ((1 - \varepsilon)(1 - \alpha) - \tau)K := \rho_{\text{eff}} K$$

for some $\rho_{\text{eff}} > 1/2$.

Define G_{eff} to be any subset of indices with $I_g = 1$ for all $g \in G_{\text{eff}}$ and $|G_{\text{eff}}| = \sum_{r=1}^K I_r$. By construction, on $\mathcal{E}_G$ we have

$$\begin{aligned} \|p_g - p^*\|_2 &\leq R_G(\alpha) = \varepsilon_{\text{eff}}, \quad g \in G_{\text{eff}}, \\ |G_{\text{eff}}| &\geq \rho_{\text{eff}} K. \end{aligned} \quad (29)$$

so (29) holds.

Step 2: Effective bad cluster. Consider

$$X_B(p) := \|p - p^*\|_2^2$$

for $p \sim \mathcal{D}_B$. By (15),

$$\mathbb{E}_{\mathcal{D}_B}[X_B] \geq \sigma^2 + \Delta^2, \quad (30)$$

and by (11), we have $0 \leq X_B(p) \leq 2$ almost surely.

Fix the threshold

$$a := \sigma^2 + \frac{\Delta^2}{2}. \quad (31)$$

From (11) and $\mathbb{E}_{\mathcal{D}_B}[X_B] \leq 2$, it follows that $\sigma^2 + \Delta^2 \leq 2$, hence $a \leq \sigma^2 + \Delta^2 \leq 2$ and in particular $a \leq 2$. Decompose

$$\begin{aligned}\mathbb{E}_{\mathcal{D}_B}[X_B] &= \mathbb{E}_{\mathcal{D}_B}[X_B \mathbf{1}\{X_B < a\}] \\ &\quad + \mathbb{E}_{\mathcal{D}_B}[X_B \mathbf{1}\{X_B \geq a\}] \\ &\leq a \cdot \Pr(X_B < a) + 2 \cdot \Pr(X_B \geq a) \\ &= a + (2 - a) \Pr(X_B \geq a)\end{aligned}\quad (32)$$

since $X_B \leq 2$ almost surely. Combining this with $\mathbb{E}_{\mathcal{D}_B}[X_B] \geq \sigma^2 + \Delta^2$ yields

$$\begin{aligned}\sigma^2 + \Delta^2 &\leq a + (2 - a) \Pr(X_B \geq a) \\ &= \sigma^2 + \frac{\Delta^2}{2} + (2 - a) \Pr(X_B \geq a),\end{aligned}\quad (33)$$

and hence

$$\Pr(X_B \geq a) \geq \frac{\frac{\Delta^2}{2}}{2 - a} \geq \frac{\Delta^2}{4}.\quad (34)$$

Equivalently,

$$\Pr_{p \sim \mathcal{D}_B} \left(\|p - p^*\|_2 \geq \sqrt{a} \right) \geq \frac{\Delta^2}{4}.\quad (35)$$

Define

$$R_B := \sqrt{a} = \sqrt{\sigma^2 + \frac{\Delta^2}{2}}, \quad \Delta_{\text{eff}} := R_B.\quad (36)$$

By Assumption (17), we have $\Delta_{\text{eff}} = R_B > R_G(\alpha) = \varepsilon_{\text{eff}}$.

Now consider the mixture $\mathcal{D}$. The probability that $p \sim \mathcal{D}$ is drawn from $\mathcal{D}_B$ and satisfies $\|p - p^*\|_2 \geq R_B$ is at least

$$\Pr_{p \sim \mathcal{D}} (p \text{ from } \mathcal{D}_B, \|p - p^*\|_2 \geq R_B) \geq \varepsilon \cdot \frac{\Delta^2}{4}.\quad (37)$$

For each r , define the indicator

$$J_r := \mathbf{1}\{p_r \text{ is drawn from } \mathcal{D}_B \text{ and } \|p_r - p^*\|_2 \geq R_B\}.$$

Then $(J_r)_{r=1}^K$ are i.i.d. Bernoulli random variables with

$$\mathbb{E}[J_r] = \Pr_{p_r \sim \mathcal{D}}(J_r = 1) \geq \varepsilon \cdot \frac{\Delta^2}{4}.\quad (38)$$

Applying Hoeffding's inequality again, for any $\tau' > 0$,

$$\Pr \left(\frac{1}{K} \sum_{r=1}^K J_r \leq \varepsilon \frac{\Delta^2}{4} - \tau' \right) \leq \exp(-2K\tau'^2).\quad (39)$$

Given δ , we may choose $\tau' > 0$ and K large enough so that $\varepsilon \frac{\Delta^2}{4} - \tau' > 0$ and $\exp(-2K\tau'^2) \leq \delta/3$.

Define the event

$$\mathcal{E}_B := \left\{ \frac{1}{K} \sum_{r=1}^K J_r > \varepsilon \frac{\Delta^2}{4} - \tau' \right\}.$$

Then $\Pr(\mathcal{E}_B) \geq 1 - \delta/3$, and on $\mathcal{E}_B$ there are at least

$$\left(\varepsilon \frac{\Delta^2}{4} - \tau' \right) K$$

indices r such that $J_r = 1$. Define B_{eff} to be any subset of indices with $J_b = 1$ for all $b \in B_{\text{eff}}$ and $|B_{\text{eff}}| = \sum_{r=1}^K J_r$. By construction, for all $b \in B_{\text{eff}}$ we have $\|p_b - p^*\|_2 \geq R_B = \Delta_{\text{eff}}$, so (22) holds on $\mathcal{E}_B$.

Step 3: Control of the barycenter. Let $\mu := \mathbb{E}[p_r]$ be the mean of the mixture $\mathcal{D}$. From (13) we have

$$\mu = (1 - \varepsilon)\mu_G + \varepsilon\mu_B.\quad (40)$$

Using Jensen's inequality and (14),

$$\|\mu_G - p^*\|_2^2 \leq \mathbb{E}_{\mathcal{D}_G}[\|p - p^*\|_2^2] \leq \sigma^2,\quad (41)$$

so $\|\mu_G - p^*\|_2 \leq \sigma$. Hence

$$\begin{aligned}\|\mu - p^*\|_2 &= \|(1 - \varepsilon)(\mu_G - p^*) + \varepsilon(\mu_B - p^*)\|_2 \\ &\leq (1 - \varepsilon)\|\mu_G - p^*\|_2 + \varepsilon\|\mu_B - p^*\|_2 \\ &\leq \|\mu_G - p^*\|_2 + \varepsilon\|\mu_B - \mu_G\|_2 \\ &\leq \sigma + \varepsilon\|\mu_B - \mu_G\|_2 \\ &\leq \sigma + c_0\Delta,\end{aligned}\quad (42)$$

where we used (16) in the last inequality.

Now consider the empirical barycenter $\bar{p} = \frac{1}{K} \sum_{r=1}^K p_r$. Since each $p_r \in \Delta^{V-1}$ with $\|p_r\|_2 \leq 1$, the vector-valued Hoeffding inequality implies that, for any $t > 0$,

$$\Pr(\|\bar{p} - \mu\|_2 \geq t) \leq 2 \exp(-cKt^2),\quad (43)$$

for some universal constant $c > 0$. Given δ , choose $t > 0$ and K large enough such that $2 \exp(-cKt^2) \leq \delta/3$. Define

$$\mathcal{E}_M := \{\|\bar{p} - \mu\|_2 \leq t\}.$$

Then $\Pr(\mathcal{E}_M) \geq 1 - \delta/3$, and on $\mathcal{E}_M$,

$$\|\bar{p} - p^*\|_2 \leq \|\bar{p} - \mu\|_2 + \|\mu - p^*\|_2 \leq t + \sigma + c_0\Delta.\quad (44)$$

We now ensure that this is bounded by a fraction of the gap $\Delta_{\text{eff}} - \varepsilon_{\text{eff}} = R_B - R_G(\alpha) > 0$. By

Assumption (17), $R_G(\alpha) < R_B$, so $\Delta_{\text{eff}} - \varepsilon_{\text{eff}} > 0$. Fix any $\eta \in (0, \frac{1}{2})$. By increasing K , we can make t arbitrarily small, and therefore we can choose K so large that

$$t + \sigma + c_0 \Delta \leq \eta(R_B - R_G(\alpha)) = \eta(\Delta_{\text{eff}} - \varepsilon_{\text{eff}}). \quad (45)$$

On $\mathcal{E}_M$ we then have

$$\|\bar{p} - p^*\|_2 \leq \eta(\Delta_{\text{eff}} - \varepsilon_{\text{eff}}),$$

which is (23).

Step 4: Union bound. Define

$$\mathcal{E} := \mathcal{E}_G \cap \mathcal{E}_B \cap \mathcal{E}_M.$$

By construction and our choices of K , we have

$$\Pr(\mathcal{E}) \geq 1 - \left(\frac{\delta}{3} + \frac{\delta}{3} + \frac{\delta}{3}\right) = 1 - \delta,$$

and on $\mathcal{E}$ all three properties hold. This proves the lemma. $\square$

Lemma 1 states that, for sufficiently many rollouts, with high probability the empirical set behaves as if there were a deterministic “good cluster” and “bad cluster” around p^* , with the barycenter $\bar{p}$ staying within the attraction region of the good cluster. We next exploit this for robust aggregation.

G.3 Robust Aggregation via Squared ℓ_2

We now show that, on the high-probability event of Lemma 1, exponential weighting based on the squared ℓ_2 distance D_r suppresses contaminated rollouts exponentially.

For the moment, we ignore the confidence term ($\lambda = 0$) and consider pure distance-based weights

$$w_r \propto \exp(-\beta D_r), D_r = \frac{1}{2} \|p_r - \bar{p}\|_2^2, \quad (46)$$

Theorem 2 (Robust aggregation under geometric separation). *Suppose the high-probability event of Lemma 1 holds, with parameters $\varepsilon_{\text{eff}}, \Delta_{\text{eff}}, \rho_{\text{eff}}, \eta$ satisfying $\Delta_{\text{eff}} > \varepsilon_{\text{eff}}$ and $\eta < \frac{1}{2}$. Then there exists a constant $\gamma_{\text{eff}} > 0$, depending only on these parameters, such that:*

1. For all $g \in G_{\text{eff}}$ and $b \in B_{\text{eff}}$,

$$D_b \geq D_g + \gamma_{\text{eff}}. \quad (47)$$

2. For any $\beta > 0$, the aggregate distribution $q = \sum_{r=1}^K w_r p_r$ with $w_r \propto \exp(-\beta D_r)$ satisfies

$$\|q - p^*\|_2 \leq \varepsilon_{\text{eff}} + C_U + \frac{1 - \rho_{\text{eff}}}{\rho_{\text{eff}}} e^{-\beta \gamma_{\text{eff}}} \quad (48)$$

where C_U is a constant. In particular, if $G_{\text{eff}} \cup B_{\text{eff}} = [K]$, the aggregated distribution q converges in ℓ_2 to the effective good cluster up to radius ε_{eff} , and the influence of contaminated rollouts is exponentially suppressed.

Proof. Step 1: Gap in D_r . By Lemma 1, for all $g \in G_{\text{eff}}$ we have $\|p_g - p^*\|_2 \leq \varepsilon_{\text{eff}}$ and for all $b \in B_{\text{eff}}$ we have $\|p_b - p^*\|_2 \geq \Delta_{\text{eff}}$, and the barycenter satisfies $\|\bar{p} - p^*\|_2 \leq \eta(\Delta_{\text{eff}} - \varepsilon_{\text{eff}})$.

For any $g \in G_{\text{eff}}$,

$$\begin{aligned} \|p_g - \bar{p}\|_2 &\leq \|p_g - p^*\|_2 + \|p^* - \bar{p}\|_2 \\ &\leq \varepsilon_{\text{eff}} + \eta(\Delta_{\text{eff}} - \varepsilon_{\text{eff}}), \end{aligned} \quad (49)$$

so

$$\begin{aligned} D_g &= \frac{1}{2} \|p_g - \bar{p}\|_2^2 \leq \frac{1}{2} (\varepsilon_{\text{eff}} + \eta(\Delta_{\text{eff}} - \varepsilon_{\text{eff}}))^2 \\ &=: D_g^{\max}. \end{aligned} \quad (50)$$

Similarly, for any $b \in B_{\text{eff}}$,

$$\begin{aligned} \|p_b - \bar{p}\|_2 &\geq \left| \|p_b - p^*\|_2 - \|p^* - \bar{p}\|_2 \right| \\ &\geq \Delta_{\text{eff}} - \eta(\Delta_{\text{eff}} - \varepsilon_{\text{eff}}), \end{aligned} \quad (51)$$

and thus

$$\begin{aligned} D_b &= \frac{1}{2} \|p_b - \bar{p}\|_2^2 \geq \frac{1}{2} (\Delta_{\text{eff}} - \eta(\Delta_{\text{eff}} - \varepsilon_{\text{eff}}))^2 \\ &=: D_b^{\min}. \end{aligned} \quad (52)$$

Define

$$f(\eta) := D_b^{\min} - D_g^{\max}$$

At $\eta = 0$ we have

$$f(0) = \frac{1}{2} (\Delta_{\text{eff}}^2 - \varepsilon_{\text{eff}}^2) > 0$$

since $\Delta_{\text{eff}} > \varepsilon_{\text{eff}}$. The map $\eta \mapsto f(\eta)$ is continuous on $[0, \frac{1}{2})$, so there exists $\eta_0 \in (0, \frac{1}{2})$ such that $f(\eta) > 0$ for all $\eta \in [0, \eta_0]$. Lemma 1 guarantees that η can be chosen in $(0, \frac{1}{2})$; by further shrinking η if necessary we may assume $\eta \leq \eta_0$. Define

$$\gamma_{\text{eff}} := f(\eta) > 0. \quad (53)$$

It follows that, for all $g \in G_{\text{eff}}$ and $b \in B_{\text{eff}}$,

$$D_b \geq D_b^{\min} = D_g^{\max} + \gamma_{\text{eff}} \geq D_g + \gamma_{\text{eff}},$$

which proves (47).

Step 2: Exponential suppression and error bound. Define the remaining index set

$$U_{\text{eff}} := [K] \setminus (G_{\text{eff}} \cup B_{\text{eff}}),$$

and the corresponding total weights

$$\begin{aligned} W_B &:= \sum_{b \in B_{\text{eff}}} w_b, W_G := \sum_{g \in G_{\text{eff}}} w_g, \\ W_U &:= \sum_{u \in U_{\text{eff}}} w_u, \end{aligned} \quad (54)$$

so that $W_B + W_G + W_U = 1$.

Let

$$\begin{aligned} A &:= \sum_{g \in G_{\text{eff}}} e^{-\beta D_g}, B := \sum_{b \in B_{\text{eff}}} e^{-\beta D_b}, \\ C &:= \sum_{u \in U_{\text{eff}}} e^{-\beta D_u}, Z := A + B + C. \end{aligned} \quad (55)$$

Then for every $r \in [K]$,

$$w_r = \frac{e^{-\beta D_r}}{Z}, \quad \text{and} \quad W_B = \frac{B}{Z}.$$

Using (47), for any $b \in B_{\text{eff}}$ and any $g \in G_{\text{eff}}$,

$$e^{-\beta D_b} \leq e^{-\beta(D_g + \gamma_{\text{eff}})} = e^{-\beta \gamma_{\text{eff}}} e^{-\beta D_g}.$$

Taking min over g and summing over b gives

$$B \leq |B_{\text{eff}}| e^{-\beta \gamma_{\text{eff}}} \min_{g \in G_{\text{eff}}} e^{-\beta D_g} \quad (56)$$

$$\leq |B_{\text{eff}}| e^{-\beta \gamma_{\text{eff}}} \frac{1}{|G_{\text{eff}}|} \sum_{g \in G_{\text{eff}}} e^{-\beta D_g} \quad (57)$$

$$= \frac{|B_{\text{eff}}|}{|G_{\text{eff}}|} e^{-\beta \gamma_{\text{eff}}} A, \quad (58)$$

and thus

$$R := \frac{B}{A} \leq \frac{|B_{\text{eff}}|}{|G_{\text{eff}}|} e^{-\beta \gamma_{\text{eff}}}.$$

Since $|G_{\text{eff}}| \geq \rho_{\text{eff}} K$ and $|B_{\text{eff}}| \leq K - |G_{\text{eff}}| \leq (1 - \rho_{\text{eff}})K$, we obtain

$$R \leq \frac{1 - \rho_{\text{eff}}}{\rho_{\text{eff}}} e^{-\beta \gamma_{\text{eff}}}.$$

Moreover, because $Z \geq A + B$,

$$W_B = \frac{B}{Z} \leq \frac{B}{A + B} = \frac{R}{1 + R} \leq R,$$

so

$$W_B \leq \frac{1 - \rho_{\text{eff}}}{\rho_{\text{eff}}} e^{-\beta \gamma_{\text{eff}}}.$$

Finally,

$$\begin{aligned} \|q - p^*\|_2 &= \left\| \sum_{r=1}^K w_r (p_r - p^*) \right\|_2 \\ &\leq \sum_{r=1}^K w_r \|p_r - p^*\|_2 \\ &\leq \varepsilon_{\text{eff}} \sum_{g \in G_{\text{eff}}} w_g + \Delta_{\text{max}} \sum_{r \notin G_{\text{eff}}} w_r \\ &= \varepsilon_{\text{eff}} W_G + \Delta_{\text{max}} (W_B + W_U), \end{aligned} \quad (59)$$

where $\Delta_{\text{max}} := \max_{1 \leq r \leq K} \|p_r - p^*\|_2 \leq \sqrt{2}$ for distributions on the simplex.

Using $W_G = 1 - W_B - W_U$, (59) implies

$$\begin{aligned} \|q - p^*\|_2 &\leq \varepsilon_{\text{eff}} (1 - W_B - W_U) + \\ &\quad \Delta_{\text{max}} (W_B + W_U) \\ &= \varepsilon_{\text{eff}} + (\Delta_{\text{max}} - \varepsilon_{\text{eff}}) (W_B + W_U) \\ &\leq \varepsilon_{\text{eff}} + (\Delta_{\text{max}} - \varepsilon_{\text{eff}}) W_U + \\ &\quad (\Delta_{\text{max}} - \varepsilon_{\text{eff}}) \frac{1 - \rho_{\text{eff}}}{\rho_{\text{eff}}} e^{-\beta \gamma_{\text{eff}}}. \end{aligned} \quad (60)$$

Defining the residual term

$$C_U := (\Delta_{\text{max}} - \varepsilon_{\text{eff}}) W_U \quad (\leq \Delta_{\text{max}} - \varepsilon_{\text{eff}}),$$

we can rewrite (60) in the same final form as

$$\|q - p^*\|_2 \leq \varepsilon_{\text{eff}} + C_U + (\Delta_{\text{max}} - \varepsilon_{\text{eff}}) \frac{1 - \rho_{\text{eff}}}{\rho_{\text{eff}}} e^{-\beta \gamma_{\text{eff}}}.$$

□

G.4 Effect of the Margin Term as a Bounded Perturbation

We now return to the full weighting scheme, which includes a margin-based confidence term $C_r \in [0, 1]$:

$$s_r = \lambda C_r - \beta D_r, \quad w_r \propto \exp(s_r). \quad (61)$$

Since $C_r \in [0, 1]$, the margin term perturbs each log-weight by at most λ :

$$\begin{aligned} -\beta D_r &\leq s_r \leq -\beta D_r + \lambda \Rightarrow \\ e^{-\beta D_r} &\leq e^{s_r} \leq e^\lambda e^{-\beta D_r}. \end{aligned} \quad (62)$$

Corollary 1 (Robustness with margin-based confidence). *Under the high-probability event of Lemma 1, consider the full weighting scheme*

$$\begin{aligned} w_r &\propto \exp(\lambda C_r - \beta D_r), \quad C_r \in [0, 1], \\ D_r &= \frac{1}{2} \|p_r - \bar{p}\|_2^2. \end{aligned} \quad (63)$$

Let $U_{\text{eff}} := [K] \setminus (G_{\text{eff}} \cup B_{\text{eff}})$ and

$$W_U := \sum_{u \in U_{\text{eff}}} w_u.$$

Let $\Delta_{\text{max}} := \max_{1 \leq r \leq K} \|p_r - p^*\|_2$ (for distributions on the simplex, $\Delta_{\text{max}} \leq \sqrt{2}$). Then, for any $\beta > 0$,

$$\begin{aligned} \|q - p^*\|_2 &\leq \varepsilon_{\text{eff}} + (\Delta_{\text{max}} - \varepsilon_{\text{eff}}) W_U + \\ &\quad (\Delta_{\text{max}} - \varepsilon_{\text{eff}}) \frac{1 - \rho_{\text{eff}}}{\rho_{\text{eff}}} \exp(-\beta \gamma_{\text{eff}} + \lambda). \end{aligned} \quad (64)$$

In particular, as long as $\beta \gamma_{\text{eff}} > \lambda$, the influence of B_{eff} is exponentially suppressed (up to constant factors).

Proof. Let w_r be the full weights with $s_r = \lambda C_r - \beta D_r$. Define the (unnormalized) sums

$$\begin{aligned} A_s &:= \sum_{g \in G_{\text{eff}}} e^{s_g}, \quad B_s := \sum_{b \in B_{\text{eff}}} e^{s_b}, \\ C_s &:= \sum_{u \in U_{\text{eff}}} e^{s_u}, \quad Z_s := A_s + B_s + C_s. \end{aligned} \quad (65)$$

Then $w_r = e^{s_r}/Z_s$ and $W_B := \sum_{b \in B_{\text{eff}}} w_b = B_s/Z_s$. For any $b \in B_{\text{eff}}$ and $g \in G_{\text{eff}}$, using $C_b \leq 1$, $C_g \geq 0$ and (47),

$$s_b - s_g = \lambda(C_b - C_g) - \beta(D_b - D_g) \leq \lambda - \beta\gamma_{\text{eff}},$$

hence

$$e^{s_b} \leq \exp(-\beta\gamma_{\text{eff}} + \lambda) e^{s_g}.$$

Taking min over g and summing over b yields

$$\begin{aligned} B_s &\leq |B_{\text{eff}}| \exp(-\beta\gamma_{\text{eff}} + \lambda) \min_{g \in G_{\text{eff}}} e^{s_g} \\ &\leq \frac{|B_{\text{eff}}|}{|G_{\text{eff}}|} \exp(-\beta\gamma_{\text{eff}} + \lambda) \sum_{g \in G_{\text{eff}}} e^{s_g} \\ &= \frac{|B_{\text{eff}}|}{|G_{\text{eff}}|} \exp(-\beta\gamma_{\text{eff}} + \lambda) A_s. \end{aligned} \quad (66)$$

Therefore, with $R_s := B_s/A_s$,

$$\begin{aligned} R_s &\leq \frac{|B_{\text{eff}}|}{|G_{\text{eff}}|} \exp(-\beta\gamma_{\text{eff}} + \lambda) \\ &\leq \frac{1 - \rho_{\text{eff}}}{\rho_{\text{eff}}} \exp(-\beta\gamma_{\text{eff}} + \lambda), \end{aligned} \quad (67)$$

where we used $|G_{\text{eff}}| \geq \rho_{\text{eff}} K$ and $|B_{\text{eff}}| \leq K - |G_{\text{eff}}| \leq (1 - \rho_{\text{eff}})K$. Moreover, since $Z_s \geq A_s + B_s$,

$$\begin{aligned} W_B &= \frac{B_s}{Z_s} \leq \frac{B_s}{A_s + B_s} = \frac{R_s}{1 + R_s} \leq R_s \\ &\leq \frac{1 - \rho_{\text{eff}}}{\rho_{\text{eff}}} \exp(-\beta\gamma_{\text{eff}} + \lambda). \end{aligned} \quad (68)$$

Finally, define

$$W_G := \sum_{g \in G_{\text{eff}}} w_g, \quad W_U := \sum_{u \in U_{\text{eff}}} w_u,$$

so $W_G + W_B + W_U = 1$. By the same triangle-inequality argument as in the robust-aggregation proof,

$$\begin{aligned} \|q - p^*\|_2 &\leq \varepsilon_{\text{eff}} W_G + \Delta_{\text{max}}(W_B + W_U) \\ &= \varepsilon_{\text{eff}} + (\Delta_{\text{max}} - \varepsilon_{\text{eff}})(W_B + W_U). \end{aligned} \quad (69)$$

Plugging in the bound on W_B gives (64). $\square$

H Human Annotation and Ethical Considerations

This appendix reports the human-in-the-loop procedures used in our study. All human involvement in this work concerns expert *evaluation* and *revision* of model-generated drafts, and does not involve any new patient data collection.

H.1 Instructions Given to Participants

H.1.1 Quality Assessment of Model-Generated Drafts

We ask dermatology experts to review a **900-case core set** and rate the quality of Gemini-generated initial drafts.

Instruction. Please review the provided dermatology image and the corresponding AI-generated report. Using a 0–5 Likert scale, rate the following two dimensions:

- **Morphological Fidelity:** Are the described clinical features (e.g., color, border, lesion type) fully consistent with the visual evidence in the image?
- **Reasoning Validity:** Is the chain-of-thought reasoning logically sound and properly grounded in visual evidence from the image?

Score definition. 5 indicates fully accurate and logically rigorous; 0 indicates severe errors such as major misdiagnosis or hallucinated features.

H.1.2 Gold Standard Manual Revision for the Core Set

Experts revise model-generated drafts using a dedicated web interface.

Instruction. The text box contains an AI-generated draft. Please perform the following:

1. **Line-by-line revision:** Compare against the original image and manually correct terminology errors, missing key features, or reasoning gaps.
2. **Bottleneck verification:** Ensure the revised <morph> JSON strictly follows the Derm7pt/SkinCon schema.
3. **Final approval:** The revised content should represent the clinical *gold-standard* answer for this case.

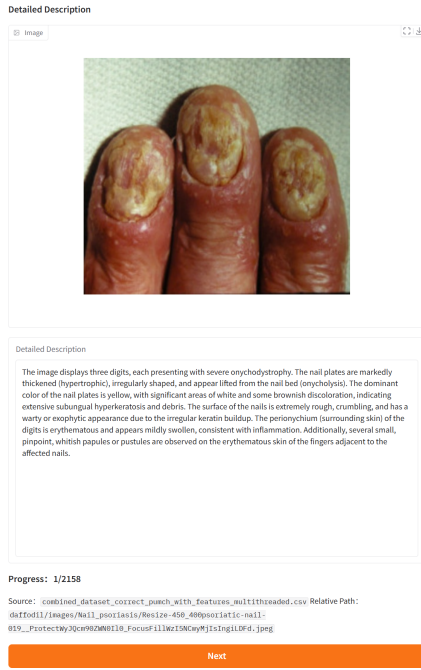


Figure 7: An example of web interface used to get .

H.1.3 Human Sanity Check for LLM-as-a-Judge

For 20 randomly sampled cases, experts evaluate whether the Judge (Gemini-2.5-Pro) provides reasonable scores and feedback.

Instruction. Please review the **model output**, **reference answer**, and the **AI Judge’s** score and feedback.

- **Task:** Rate (0–5) whether the AI Judge’s evaluation is reasonable.
- **Reasonableness criteria:** The score should be objective, and the feedback should point out key medical differences.
- **Acceptance threshold:** Scores ≥ 3 are considered acceptable.

H.1.4 Human Performance Baseline

To obtain the “Human Performance” results, we randomly sample **100 cases per task** and ask experts to complete the benchmark **without any AI assistance**.

Instruction. Please independently complete DermoBench evaluation tasks as in clinical practice, **without referencing any AI hints**:

1. **MCQA tasks:** Select the most likely diagnosis from 4-choice or 25-choice options.

2. **Hierarchical diagnosis:** Perform step-wise selection along the diagnosis tree path (Super-class $\rightarrow$ Subclass).
3. **Open-ended description:** Write a detailed morphological examination report without viewing any reference answer.

H.2 Recruitment, Compensation, and Consent

Recruitment and qualifications. We invited and engaged two dermatology clinicians via targeted online outreach. Both participants have relevant clinical experience in dermatology.

Compensation. Participants were compensated at approximately 100 RMB per hour, following local norms for medical professional consulting, which we consider adequate to reflect the value of expert labor.

Annotator consent. All participating clinicians signed an agreement acknowledging that their revision, annotation, and rating outputs would be used for open research purposes in developing and evaluating our dermatology MLLMs and benchmark.

H.3 Data Consent, Release Policy, and Ethics Review

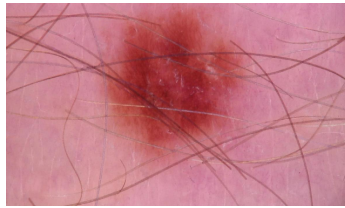
Open datasets and intended use. This work uses only publicly released, de-identified dermatology datasets. We follow the licenses and intended research use specified by the original dataset providers. Experts may view the original images during evaluation and revision; however, **we do not redistribute or release the original images**. We release only derived artifacts (e.g., prompts, annotations, benchmark splits, and evaluation outputs), and users should obtain images from the original sources.

Ethics review. We do not collect any new patient data and only use de-identified, publicly available datasets; the expert annotation activities are minimal-risk. Therefore, ethics board approval was not required under our institutional policy.

I Visualization

The following pictures provide additional visualizations and qualitative case studies to better understand the data characteristics of DermoBench.

Task 1.1 Detailed Description



Q: Summarize all important morphological features visible in this image.

GT: The dermoscopic image displays a centrally located, reddish-brown lesion with a relatively uniform, diffuse pigmentation spread throughout its body. The lesion exhibits an overall oval shape with borders that are somewhat ill-defined yet appear regular. A distinct pigment network is absent. No definite vascular structures are identified; the reddish hue appears as structureless pigmentation. No blue-whitish veil is present. There are no radiating streaks or distinct dots and globules of pigment. Furthermore, no regression structures, such as blue or white areas, are identified.

Gemini 2.5 Flash:

Based on the provided dermoscopic images, here is a summary of the important morphological features of the skin lesion:\n\n* **Overall Structure and Symmetry:** The lesion is relatively symmetrical in its round-to-oval shape but shows asymmetry in its internal structures and color distribution.\n* **Borders:** The borders are ill-defined and fuzzy, gradually fading into the surrounding skin rather than having a sharp, distinct edge.\n* **Color:** The lesion is multicolored (polychromatic), exhibiting shades of light brown, dark brown, and a prominent reddish-pink hue throughout.\n* **Central Features:**\n* **Blotch:** There is a central, darker, structureless blotch that is dark brown to reddish.\n* **Regression Structures:** Shiny white lines (also known as chrysalis structures or crystalline structures) are visible in the central area, which are signs of fibrosis or regression.\n* **Pigment Network:** A faint, atypical, and disorganized pigment network can be observed, particularly towards the periphery.\n* **Vascular Pattern:** A diffuse reddish \"blush\" or erythema is present across the lesion, indicating a significant vascular component, though specific vessel patterns are not clearly defined.\n\nIn summary, the key features are the multicomponent pattern with structural and color asymmetry, ill-defined borders, multiple colors including a prominent red hue, a central blotch, and signs of regression (shiny white lines).

LLM-as-a-Judge:

```
{\"score\":{\"final_overall\":8.8,\"rubric\":{\"accuracy\":0.42,\"completeness\":0.29,\"consistency\":0.42},\"counts\":{\"supported\":5,\"partial\":0,\"contradicted\":7,\"missing\":5,\"vague\":0,\"extra_incorrect\":0,\"total_ref_claims\":17},\"short_feedback\":\"The candidate description directly contradicts the reference on multiple key dermoscopic features, including color, symmetry, pigment network, and regression structures. It also invents findings (e.g., shiny white lines) not present in the reference text.\"}}
```

Lingshu-32B:

The image shows a close-up view of a skin lesion. The lesion appears to be a small, reddish area on the skin, surrounded by fine hairs. The lesion has a slightly raised, irregular shape and appears to be inflamed or irritated. The surrounding skin appears to be relatively normal, with no other visible abnormalities.

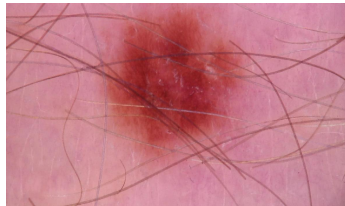
LLM-as-a-Judge:

```
{\"score\":{\"final_overall\":0.0,\"rubric\":{\"accuracy\":0.2,\"completeness\":0.1,\"consistency\":0.3},\"counts\":{\"supported\":0,\"partial\":1,\"contradicted\":2,\"missing\":14,\"vague\":1,\"extra_incorrect\":2,\"total_ref_claims\":18},\"short_feedback\":\"The candidate description is extremely superficial, missing all specific dermoscopic features. It contradicts the reference on shape and border regularity and invents findings like elevation and inflammation, resulting in a score of zero.\"}}
```

Figure 8: Case study.

Task 1.1

Detailed Description



Q: Summarize all important morphological features visible in this image.

GT: The dermoscopic image displays a centrally located, reddish-brown lesion with a relatively uniform, diffuse pigmentation spread throughout its body. The lesion exhibits an overall oval shape with borders that are somewhat ill-defined yet appear regular. A distinct pigment network is absent. No definite vascular structures are identified; the reddish hue appears as structureless pigmentation. No blue-whitish veil is present. There are no radiating streaks or distinct dots and globules of pigment. Furthermore, no regression structures, such as blue or white areas, are identified.

DermoGPT-RL:

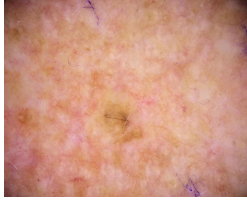
Dermoscopy reveals a relatively symmetrical, round to oval lesion with a predominantly reddish-brown coloration. The pigmentation is diffuse and appears largely regular throughout the lesion, without significant areas of darker blotches or lighter patches. A distinct pigment network is not observed. There is no evidence of a blue-whitish veil. While the lesion has a reddish hue, no specific vascular structures such as arborizing, comma, hairpin, wreath, dotted, or linear irregular vessels are clearly identifiable. The lesion lacks any radiating streaks or pseudopods at its periphery. Furthermore, no discrete dots or globules are present within the lesion. There are no signs of regression, such as blue areas or white scar-like areas.

LLM-as-a-Judge:

```
{"score":{"final_overall":88.2,"rubric":{"accuracy":1.0,"completeness":0.941,"consistency":1.0},"counts":{"supported":14,"partial":2,"contradicted":0,"missing":1,"vague":0,"extra_incorrect":0,"total_ref_claims":17},"short_feedback":"Excellent accuracy and consistency. The candidate is slightly verbose but captures nearly all key features. The only missing detail is the 'ill-defined' nature of the borders, a minor omission in an otherwise comprehensive description."}}
```

Figure 9: Case study.

Task 1.2 Morph-grounded Description



Q: Describe the lesion based on the 7-point checklist (pigment network, blue-whitish veil, vascular structures, pigmentation, streaks, dots/globules, and regression structures).
Before any reasoning, first output a <morph> JSON using the Derm7pt schema, then provide exactly ONE detailed paragraph."

GT: <morph>\n{\n \"morphological_features_Derm7pt\": {\n \"pigment_network\": \"absent\", \n \"blue_whitish_veil\": \"absent\", \n \"vascular_structures\": \"linear irregular\", \n \"pigmentation\": \"diffuse irregular\", \n \"streaks\": \"absent\", \n \"dots_and_globules\": \"absent\", \n \"regression_structures\": \"absent\" \n } \n } \n</morph>\n\nDermoscopy of this skin area reveals a broad, ill-defined lesion with diffuse and irregular pigmentation, primarily consisting of light brown and yellowish-brown patches that blend into the surrounding skin without clear demarcation. A pigment network is not distinctly discernible within this variegated background. Numerous fine, linear irregular vascular structures are observed, distributed somewhat haphazardly across the lesion, contributing to a subtle background erythema. There is no evidence of blue-whitish veil, streaks, or distinct dots and globules that would classify as regular or irregular. Furthermore, no regression structures, such as blue areas or white areas, are present.

Gemini 2.5 Flash:

<morph>\n{\n \"morphological_features_Derm7pt\": {\n \"pigment_network\": \"atypical\", \n \"blue_whitish_veil\": \"absent\", \n \"vascular_structures\": \"linear irregular\", \n \"pigmentation\": \"diffuse irregular\", \n \"streaks\": \"absent\", \n \"dots_and_globules\": \"irregular\", \n \"regression_structures\": \"white areas\" \n } \n } \n\nThis lesion is characterized by diffuse irregular pigmentation, with blotchy areas of tan and light brown coloration distributed unevenly over a pinkish background. A faint and focally present atypical pigment network is visible, appearing as broad and interrupted lines. The vascular pattern consists of multiple fine, short, linear irregular vessels scattered throughout the lesion. No blue-whitish veil or peripheral streaks are observed.

LLM-as-a-Judge:

{\"score\":{\"final_overall\":39.85,\"rubric\":{\"accuracy\":0.7,\"completeness\":0.47,\"consistency\":1.0},\"counts\":{\"supported\":7,\"partial\":0,\"contradicted\":3,\"missing\":5,\"vague\":0,\"extra_incorrect\":0,\"total_ref_claims\":15},\"short_feedback\":\"The candidate narrative correctly identifies several features but **contradicts the reference on key findings** like pigment network, dots/globules, and regression. It also misses describing the lesion's broad, ill-defined nature.\"}

Lingshu-32B:

<morph>\n{\n \"morphological_features_Derm7pt\": {\n \"pigment_network\": \"absent\", \n \"blue_whitish_veil\": \"absent\", \n \"vascular_structures\": \"absent\", \n \"pigmentation\": \"absent\", \n \"streaks\": \"absent\", \n \"dots_and_globules\": \"absent\", \n \"regression_structures\": \"absent\" \n } \n } \n\nThe dermoscopic image shows a lesion with no discernible pigment network, indicating an absence of the typical interconnecting lines of pigment. There is no visible blue-whitish veil, which would have appeared as a whitish or bluish haze over the lesion. The vascular structures are not present, meaning there are no visible blood vessels or vascular patterns. The dots and globules are absent, indicating no small, round structures of pigment. Lastly, there are no regression structures, such as blue or white areas, which would indicate areas of regression within the lesion.

LLM-as-a-Judge:

{\"score\":{\"final_overall\":42.15,\"rubric\":{\"accuracy\":0.45,\"completeness\":0.33,\"consistency\":1.0},\"count s\":{\"supported\":5,\"partial\":0,\"contradicted\":6,\"missing\":4,\"vague\":0,\"extra_incorrect\":0,\"total_ref_clai ms\":15},\"short_feedback\":\"The candidate correctly identifies the absence of several features but **severely mischaracterizes the lesion** by stating pigmentation and vascular structures are absent.\"}

Figure 10: Case study.

Task 1.2 Morph-grounded Description



Q: Describe the lesion based on the 7-point checklist (pigment network, blue-whitish veil, vascular structures, pigmentation, streaks, dots/globules, and regression structures).
Before any reasoning, first output a <morph> JSON using the Derm7pt schema, then provide exactly ONE detailed paragraph."

GT: <morph>\n{\n \"morphological_features_Derm7pt\": {\n \"pigment_network\": \"absent\", \n \"blue_whitish_veil\": \"absent\", \n \"vascular_structures\": \"linear irregular\", \n \"pigmentation\": \"diffuse irregular\", \n \"streaks\": \"absent\", \n \"dots_and_globules\": \"absent\", \n \"regression_structures\": \"absent\" \n } \n } \n</morph>\n\nDermoscopy of this skin area reveals a broad, ill-defined lesion with diffuse and irregular pigmentation, primarily consisting of light brown and yellowish-brown patches that blend into the surrounding skin without clear demarcation. A pigment network is not distinctly discernible within this variegated background. Numerous fine, linear irregular vascular structures are observed, distributed somewhat haphazardly across the lesion, contributing to a subtle background erythema. There is no evidence of blue-whitish veil, streaks, or distinct dots and globules that would classify as regular or irregular. Furthermore, no regression structures, such as blue areas or white areas, are present.

DermoGPT-RL:

<morph>\n{\n \"morphological_features_Derm7pt\": {\n \"Pigment Network\": \"absent\", \n \"Blue Whitish Veil\": \"absent\", \n \"Vascular Structures\": \"linear irregular\", \n \"Pigmentation\": \"diffuse irregular\", \n \"Streaks\": \"absent\", \n \"Dots and Globules\": \"absent\", \n \"Regression Structures\": \"absent\" \n } \n } \n</morph>\n\nDermoscopy reveals a large, ill-defined lesion characterized by a variegated appearance with areas of light brown, yellowish-brown, and pinkish-red discoloration. The pigmentation is diffuse and irregular, lacking a clear, organized pattern. A distinct pigment network is absent. There is no evidence of a blue-whitish veil. Fine, somewhat tortuous, and irregularly distributed linear vessels are visible, particularly within the pinkish-red areas, classifying them as 'linear irregular' vascular structures. No streaks or pseudopods are observed. Discrete dots or globules are not a prominent feature, and no regression structures such as blue or white areas are present. The lesion exhibits asymmetry in its color distribution and overall morphology.

LLM-as-a-Judge:

{\"score\":{\"final_overall\":98.1,\"rubric\":{\"accuracy\":1.0,\"completeness\":0.96,\"consistency\":1.0},\"counts\":{\"supported\":12,\"partial\":1,\"contradicted\":0,\"missing\":0,\"vague\":0,\"extra_incorrect\":0,\"total_ref_claims\":13},\"short_feedback\":\"Excellent narrative that accurately supports almost all reference claims. It correctly identifies all key dermoscopic features and their absence. The description of background erythema is slightly less direct but implied.\"}

Figure 11: Case study.

Task 3.1 CoT Reasoning



Q:Based on the provided image, what is the most likely diagnosis? Please provide a detailed reasoning process before giving the final answer.

GT: <reasoning>Upon examination, the lesion presents with a largely homogeneous reddish-brown coloration, which is a reassuring feature, as significant color variegation is absent. While the borders appear somewhat ill-defined and diffuse, gradually fading into the perilesional skin, they lack the sharp irregularities, notching, or sharp demarcations that would be concerning for malignancy. There is no atypical pigment network, streaks, or blue-white veil visible. The overall uniform pigment distribution, despite the somewhat irregular contour, supports a benign diagnosis.</reasoning><final_diagnosis>Benign Lesion</final_diagnosis>

Gemini 2.5 Flash:

<reasoning>\n1. The dermoscopic image displays a solitary, flat (macular) pigmented lesion on hair-bearing skin.\n2. The lesion is markedly asymmetric in both shape and color. The borders are ill-defined and fade into the surrounding skin, particularly on the left side.\n3. The color is heterogeneous, with shades ranging from light tan to dark brown. There is a darker, more disorganized area in the center-right portion of the lesion.\n4. A subtle, disorganized, and atypical pigment network is present. More importantly, there are signs of asymmetric pigmentation around the hair follicle openings (asymmetric follicular openings), a key feature of this diagnosis.\n5. The combination of asymmetry, ill-defined borders, color variegation, and specific dermoscopic patterns like asymmetric follicular openings, especially on what appears to be sun-exposed skin, is highly indicative of a melanoma in situ on sun-damaged skin.</reasoning>\n<final_diagnosis>\nLentigo maligna\n</final_diagnosis>

LLM-as-a-Judge:

{"score":{"final_overall":5.0,"rubric":{"reasoning_alignment":0.0,"diagnosis_similarity":10.0,"internal_consistency":100.0},"counts":{"supported":1,"partial":0,"contradicted":5,"missing":3,"vague":0,"extra_in_correct":2,"total_ref_claims":9},"short_feedback":"The reasoning **directly contradicts the ground truth on multiple key features**, such as color and pigment network, leading to an incorrect diagnosis of malignancy instead of a benign lesion. The candidate's findings are internally consistent."}

Lingshu-32B:

<reasoning>The image shows a close-up view of a skin lesion with a brownish color and irregular shape. The lesion appears to have a slightly raised surface and is surrounded by hair follicles. The irregular pigmentation and asymmetry are notable features. These characteristics are commonly associated with melanocytic lesions, such as melanoma. The irregular shape and uneven color distribution are key indicators that differentiate melanoma from benign lesions like nevi. Considering the visual cues, the lesion's asymmetry, irregular borders, and uneven coloration strongly suggest a diagnosis of melanoma. Other differential diagnoses, such as seborrheic keratosis or lentigo, are less likely due to the lack of typical features such as a waxy appearance or uniform pigmentation.</reasoning>\n<final_diagnosis>Melanoma</final_diagnosis>

LLM-as-a-Judge:

{"score":{"final_overall":35.0,"rubric":{"reasoning_alignment":0.0,"diagnosis_similarity":70.0,"internal_consistency":1.0},"counts":{"supported":0,"partial":1,"contradicted":5,"missing":3,"vague":0,"extra_in_correct":2,"total_ref_claims":9},"short_feedback":"The candidate's reasoning **directly contradicts the ground truth's description of a benign lesion**, incorrectly identifying features of malignancy (irregular color/borders)."} }

Figure 12: Case study.

Task 3.1 CoT Reasoning



Q:Based on the provided image, what is the most likely diagnosis? Please provide a detailed reasoning process before giving the final answer.

GT: <reasoning>Upon examination, the lesion presents with a largely homogeneous reddish-brown coloration, which is a reassuring feature, as significant color variegation is absent. While the borders appear somewhat ill-defined and diffuse, gradually fading into the perilesional skin, they lack the sharp irregularities, notching, or sharp demarcations that would be concerning for malignancy. There is no atypical pigment network, streaks, or blue-white veil visible. The overall uniform pigment distribution, despite the somewhat irregular contour, supports a benign diagnosis.</reasoning><final_diagnosis>Benign Lesion</final_diagnosis>

DermoGPT-RL:

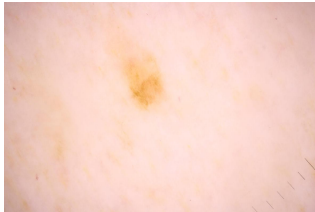
<morph>\n{\n \n "morphological_features_Derm7pt": {\n \n "Pigment Network": "\nabsent",\n \n "Blue Whitish Veil": "\nabsent",\n \n "Vascular Structures": "\nlinear irregular",\n \n "Pigmentation": "\ndiffuse irregular",\n \n "Streaks": "\nabsent",\n \n "Dots and Globules": "\nabsent",\n \n "Regression Structures": "\nabsent",\n \n }\n}\n</morph>\n\nDermoscopy reveals a large, ill-defined lesion characterized by a variegated appearance with areas of light brown, yellowish-brown, and pinkish-red discoloration. The pigmentation is diffuse and irregular, lacking a clear, organized pattern. A distinct pigment network is absent. There is no evidence of a blue-whitish veil. Fine, somewhat tortuous, and irregularly distributed linear vessels are visible, particularly within the pinkish-red areas, classifying them as 'linear irregular' vascular structures. No streaks or pseudopods are observed. Discrete dots or globules are not a prominent feature, and no regression structures such as blue or white areas are present. The lesion exhibits asymmetry in its color distribution and overall morphology.

LLM-as-a-Judge:

{"score":{"final_overall":98.1,"rubric":{"accuracy":1.0,"completeness":0.96,"consistency":1.0},"counts":{"supported":12,"partial":1,"contradicted":0,"missing":0,"vague":0,"extra_incorrect":0,"total_ref_claims":13},"short_feedback":"Excellent narrative that accurately supports almost all reference claims. It correctly identifies all key dermoscopic features and their absence. The description of background erythema is slightly less direct but implied."}

Figure 13: Case study.

Task 3.2 Morph-Grounded Reasoning



Q: Analyze the clinical presentation in this image. What is your differential diagnosis, and what is the final conclusion? Explain your reasoning. \n\n Then output EXACTLY three blocks in this order and nothing else: \n<reasoning>your step-by-step, image-grounded reasoning</reasoning> \n<morph>{STRICT JSON using the Derm7pt schema; exactly one value per field; valid JSON}</morph> \n<final_diagnosis>ONE label from our taxonomy</final_diagnosis>".

GT: <reasoning>Upon visual assessment, the lesion presents as a very subtle, flat macule. Its primary characteristic is the faint, homogeneous yellowish-brown coloration which is barely discernible from the surrounding skin. Crucially, the borders of this pigmented area are exceedingly ill-defined, blending imperceptibly into the normal skin without any sharp demarcation, notching, or irregular extensions. There is no evidence of asymmetry in shape or color distribution, nor any suspicious structures or textures suggesting an atypical melanocytic proliferation or other malignancy. The absence of variegated colors, atypical pigment network, dots, globules, streaks, or ulceration, combined with the flat, faint, and ill-defined appearance, strongly supports a benign diagnosis.</reasoning> \n<morph> \n{ \n \n "morphological_features_Derm7pt": { \n \n "pigment_network": "typical", \n \n "blue_whitish_veil": "absent", \n \n "vascular_structures": "absent", \n \n "pigmentation": "localized regular", \n \n "streaks": "absent", \n \n "dots_and_globules": "absent", \n \n "regression_structures": "absent" \n \n } \n } \n</morph> \n<final_diagnosis>Benign Lesion</final_diagnosis>

Gemini 2.5 Flash:

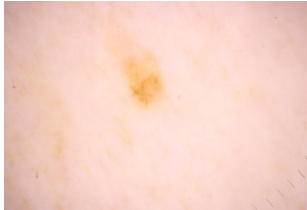
<reasoning> \n The dermoscopic image reveals a flat, poorly demarcated macule with a focal area of light brown pigmentation. \n 1. **Pigment Network**: A subtle, delicate pigment network is visible in the central pigmented area. This network is considered atypical because it is disorganized, with lines of varying thickness and irregularly sized and shaped holes. It is not the uniform, honeycomb pattern of a typical network. \n 2. **Pigmentation**: The pigmentation is localized to one part of the lesion and is irregular in its distribution and shade, varying from tan to light brown. \n 3. **Other Structures**: There is a lack of other significant dermoscopic features. Specifically, there is no blue-whitish veil, no recognizable vascular structures (like arborizing or dotted vessels), no streaks at the periphery, no prominent dots or globules, and no signs of regression (like white scar-like areas or blue-gray granules). \n 4. **Conclusion**: The key finding is the atypical pigment network within an irregularly pigmented flat lesion. This combination is a major criterion for malignancy and is highly suspicious for melanoma, particularly the lentigo maligna subtype, which often presents as a subtle, flat macule on sun-damaged skin. \n</reasoning> \n<morph> \n{ \n \n "morphological_features_Derm7pt": { \n \n "pigment_network": "atypical", \n \n "blue_whitish_veil": "absent", \n \n "vascular_structures": "absent", \n \n "pigmentation": "localized irregular", \n \n "streaks": "absent", \n \n "dots_and_globules": "absent", \n \n "regression_structures": "absent" \n \n } \n } \n</morph> \n<final_diagnosis>melanoma</final_diagnosis>

LLM-as-a-Judge:

{ "score": { "final_overall": 43.05, "rubric": { "reasoning_alignment": 7.3, "diagnosis_similarity": 50.0, "internal_consistency": 100.0 }, "counts": { "supported": 5, "partial": 0, "contradicted": 6, "missing": 0, "vague": 0, "extra_incorrect": 0, "total_ref_claims": 11 }, "short_feedback": "The candidate's reasoning and morph features are internally consistent but **directly contradict the reference's benign assessment**. The core disagreement on the pigment network and pigmentation leads to a completely different and incorrect final diagnosis of melanoma instead of a benign lesion." }

Figure 14: Case study.

Task 3.2 Morph-Grounded Reasoning



Q: Analyze the clinical presentation in this image. What is your differential diagnosis, and what is the final conclusion? Explain your reasoning. \n\nThen output EXACTLY three blocks in this order and nothing else: \n<reasoning>your step-by-step, image-grounded reasoning</reasoning> \n<morph>{STRICT JSON using the Derm7pt schema; exactly one value per field; valid JSON}</morph> \n<final_diagnosis>ONE label from our taxonomy</final_diagnosis>

GT: <reasoning>Upon visual assessment, the lesion presents as a very subtle, flat macule. Its primary characteristic is the faint, homogeneous yellowish-brown coloration which is barely discernible from the surrounding skin. Crucially, the borders of this pigmented area are exceedingly ill-defined, blending imperceptibly into the normal skin without any sharp demarcation, notching, or irregular extensions. There is no evidence of asymmetry in shape or color distribution, nor any suspicious structures or textures suggesting an atypical melanocytic proliferation or other malignancy. The absence of variegated colors, atypical pigment network, dots, globules, streaks, or ulceration, combined with the flat, faint, and ill-defined appearance, strongly supports a benign diagnosis.</reasoning> \n<morph> \n{ \n \n \"morphological_features_Derm7pt\": { \n \n \"pigment_network\": \"typical\", \n \n \"blue_whitish_veil\": \"absent\", \n \n \"vascular_structures\": \"absent\", \n \n \"pigmentation\": \"localized regular\", \n \n \"streaks\": \"absent\", \n \n \"dots_and_globules\": \"absent\", \n \n \"regression_structures\": \"absent\" \n \n } \n \n } \n</morph> \n<final_diagnosis>Benign Lesion</final_diagnosis>

Lingshu-32B:

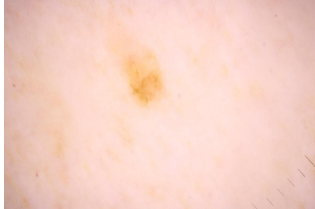
<reasoning>The image shows a light-colored lesion with a faint, slightly irregular pattern. There are no distinct vascular structures, pigment networks, or blue-whitish veils visible. The lesion appears to have a diffuse, irregular pigmentation pattern, which is not typical of benign lesions. The absence of other significant features such as streaks, dots, or globules further supports the irregularity of the lesion. Given these observations, the differential diagnosis includes conditions such as melanoma, which often presents with irregular pigmentation and lacks distinct vascular structures. The final conclusion is that the lesion is indicative of melanoma due to its irregular pigmentation and lack of other benign features.</reasoning> \n<morph> \n{ \n \n \"morphological_features_Derm7pt\": { \n \n \"pigment_network\": \"absent\", \n \n \"blue_whitish_veil\": \"absent\", \n \n \"vascular_structures\": \"absent\", \n \n \"pigmentation\": \"diffuse irregular\", \n \n \"streaks\": \"absent\", \n \n \"dots_and_globules\": \"absent\", \n \n \"regression_structures\": \"absent\" \n \n } \n \n } \n</morph> \n<final_diagnosis>melanoma</final_diagnosis>

LLM-as-a-Judge:

{\"score\":{\"final_overall\":40.64,\"rubric\":{\"reasoning_alignment\":0.0,\"diagnosis_similarity\":50.0,\"internal_consistency\":100.0},\"counts\":{\"supported\":3,\"partial\":0,\"contradicted\":6,\"missing\":0,\"vague\":1,\"extra_incorrect\":0,\"total_ref_claims\":10},\"short_feedback\":\"The candidate correctly identified the absence of most dermoscopic structures but incorrectly interpreted the overall pattern as irregular and suspicious. This led to a direct contradiction of the reference's benign assessment, resulting in an incorrect final diagnosis of melanoma.\"}}

Figure 15: Case study.

Task 3.2 Morph-Grounded Reasoning



Q: Analyze the clinical presentation in this image. What is your differential diagnosis, and what is the final conclusion? Explain your reasoning. Then output EXACTLY three blocks in this order and nothing else: `<reasoning>` your step-by-step, image-grounded reasoning `</reasoning>` `<morph>` {STRICT JSON using the Derm7pt schema; exactly one value per field; valid JSON} `</morph>` `<final_diagnosis>` ONE label from our taxonomy `</final_diagnosis>`,

GT: `<reasoning>` Upon visual assessment, the lesion presents as a very subtle, flat macule. Its primary characteristic is the faint, homogeneous yellowish-brown coloration which is barely discernible from the surrounding skin. Crucially, the borders of this pigmented area are exceedingly ill-defined, blending imperceptibly into the normal skin without any sharp demarcation, notching, or irregular extensions. There is no evidence of asymmetry in shape or color distribution, nor any suspicious structures or textures suggesting an atypical melanocytic proliferation or other malignancy. The absence of variegated colors, atypical pigment network, dots, globules, streaks, or ulceration, combined with the flat, faint, and ill-defined appearance, strongly supports a benign diagnosis. `</reasoning>` `<morph>` `{\n \"morphological_features_Derm7pt\": {\n \"pigment_network\": \"typical\", \n \"blue_whitish_veil\": \"absent\", \n \"vascular_structures\": \"absent\", \n \"pigmentation\": \"localized regular\", \n \"streaks\": \"absent\", \n \"dots_and_globules\": \"absent\", \n \"regression_structures\": \"absent\" \n } \n}` `</morph>` `<final_diagnosis>` Benign Lesion `</final_diagnosis>`

DermoGPT-RL:

`<reasoning>` Upon dermoscopic examination, the lesion presents as a macule with a predominantly yellowish-brown coloration. The pigmentation is largely homogeneous, lacking significant color variegation or atypical pigment patterns. The borders are ill-defined and gradually fade into the surrounding skin, which is a feature often associated with benign lesions such as solar lentigines or flat seborrheic keratoses. There is no evidence of asymmetry, atypical pigment network, streaks, blue-white veil, or other dermoscopic criteria suggestive of malignancy. The overall bland and uniform appearance, coupled with the ill-defined borders and yellowish-brown hue, strongly supports a benign diagnosis. `</reasoning>` `<morph>` `{\n \"morphological_features_Derm7pt\": {\n \"Pigment Network\": \"absent\", \n \"Blue Whitish Veil\": \"absent\", \n \"Vascular Structures\": \"absent\", \n \"Pigmentation\": \"localized irregular\", \n \"Streaks\": \"absent\", \n \"Dots and Globules\": \"absent\", \n \"Regression Structures\": \"absent\" \n } \n}` `</morph>` `<final_diagnosis>` Benign Lesion `</final_diagnosis>`

LLM-as-a-Judge:

{\"score\":{\"final_overall\":82.45,\"rubric\":{\"reasoning_alignment\":76.7,\"diagnosis_similarity\":100.0,\"internal_consistency\":100.0},\"counts\":{\"supported\":10,\"partial\":0,\"contradicted\":0,\"missing\":2,\"vague\":0,\"extra_incorrect\":0,\"total_ref_claims\":12},\"short_feedback\":\"Excellent reasoning and correct final diagnosis. The main area for improvement is the morph JSON, which contradicts the reference on two features (pigment network and pigmentation type), leading to a lower score in that section.\"}

Figure 16: Case study.